ECC2K-130 on Commodity Hardware: A Measured Frobenius-Class Pollard Rho Campaign

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Abstract

ECC2K-130 is the Certicom Koblitz challenge $y^2+xy = x^3+1$ over $\mathbb{F}_{2^{131}}$, open since 1997, with an expected cost of $2^{60.9}$ iterations of a Frobenius-class Pollard rho. This paper reports a measured campaign against it on commodity hardware: a type-II optimal-normal-basis field with one generated golden model, two iteration functions and a third for power-capped parts, a bitsliced CPU client, a packed CUDA client whose campaign build runs at 14.64 billion complete iterations per second on one RTX PRO 6000, reached by sixteen paired rungs from a 0.85 bitsliced control (a later σ-walk build measures 15.88, and a table walk 20.08 as a rate, not yet as a cost per solve), and a VHDL engine measured at 8.25 G steps/s on one AWS F2 FPGA against a pre-board estimate of 5–12. The repositories' other GPU arithmetic, for secp256k1, the NIST primes and ECC2K-95, is inventoried with its static instruction ledgers and emulated walk constants and the statement that, apart from F4 elimination on rented cards, none of it has yet run on a device. The walk constant is measured, not assumed: $2^{60.91}$–$2^{60.92}$ iterations for the campaign's σ-walk, so the 2009 budget stands. A carry-less-unit floor written before tuning rules out 28 B/s on one GPU, and a pre-declared target of 17.0 B/s declined a 16.56 B/s table walk. The campaign at cutoff 32, one report per $2^{28.41}$ iterations, has stored 486 million points and $2^{56.68}$ iterations, 5.4% of the expected work, with no collision; its fleet was reclaimed on 2026-09-18 and collection has been paused since 2026-10-07. Index calculus on the curve, priced in the same unit, sits 10^7 times above rho at the challenge size and the best measured point decomposition is 35–51 bits short of its budget. Every number is footnoted to a repository artefact; nothing here is an advance on the generic-group floor, and the paper says so.

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1 Introduction

In 1997 Certicom published a list of elliptic-curve discrete logarithm challenges [11]. The Koblitz challenges ECC2K-95, ECC2K-108 and ECC2K-130 ask for k with $Q = [k]P$ on $y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^{97}}$, $\mathbb{F}_{2^{109}}$ and $\mathbb{F}_{2^{131}}$. Harley's group solved ECC2K-95 in 1998 [19] and ECC2K-108 in 2000; ECC2K-130 is open. In 2009 a consortium of twenty authors published the arithmetic, the iteration function and the per-platform throughput of a planned attack [2, 3, 9, 14] and estimated the work at $2^{60.9}$ iterations. The attack did not complete.

This paper is a measured account of a second attempt, built from scratch on hardware that did not exist in 2009 and documented to a standard the first attempt's reports set: every rate paired against an adjacent control on the same allocation, every boundary written down before the kernel was tuned toward it, every claim footnoted to a file in the repository that holds it. The work is not an advance on the generic-group floor and the paper does not say it is. What it reports is the cost of one step, measured on GPUs, CPUs and an FPGA, the number of steps, measured on toy curves of the same family and extrapolated, and a campaign that has walked $2^{56.7}$ of the $2^{60.9}$, most of it before its fleet was taken away.

Contributions.

- A code generator that builds every field routine from one symmetric-vector model of $\mathbb{F}_{2^{131}}$ and checks it before emitting it; a bitsliced backend at every word width from 64 to 512 lanes and a packed CUDA backend with native carry-less products; bit-for-bit compatible records across all of them and a VHDL engine (Sections 2, 4).
- Three iteration functions on the $\langle \sigma, -1 \rangle$ classes, with the walk constant measured on toy Koblitz curves and extrapolated: $2^{60.91}$–$2^{60.92}$ iterations for the campaign's walk, with the table walk's fruitless cycles measured, priced and repaired (Section 3).
- A GPU ledger of sixteen paired rungs from 0.85 to 14.64 B/s on one RTX PRO 6000, the campaign build; later σ-walk builds at 15.44 and 15.88; a table walk at 20.08 B/s, a rate whose cost per solve is not yet measured; a twelve-SKU survey, a roofline, and a floor derived from the carry-less unit before tuning that rules out 28 B/s on one card; with every declined candidate and its ratio (Section 5).
- An inventory of the other GPU arithmetic in the repositories, prime-field (secp256k1 rho with order-six folding, baby-step giant-step, kangaroo, P-256/P-384) and binary-field (the ECC2K-95 kernel, pair tables, polynomial-system solvers), with their instruction ledgers and emulated walk constants, and a plain statement of which of them have run on a device: the F4 elimination kernels have, the rest have not (Section 7).
- A pipelined-Karatsuba FPGA engine measured at 8.25 G steps/s on one AWS F2, within a pre-board estimate of 5–12 (Section 8).
- The campaign: configuration, the calibration of the distinguished-point interval (and the retraction of a wrong one), the storage and replay protocol, fleet economics, and the state at 486 M points and $2^{56.68}$ iterations (Section 9).
- The structural alternatives priced in the same unit and found wanting: index calculus on the Koblitz curve at $10^7\times$ rho, and point decomposition 35–51 bits short of its budget (Section 10).

Method. The reporting rule of the repository, stated here once, governs every section. A change is measured as a ratio to a boundary that was written down before the change was attempted: a *floor*, from a counting, generic-group or hardware-pipe argument that the method

cannot cross, or a *reference*, the best existing method on the same instance in the same unit. Every cost the method incurs is inside the ratio: precomputation, verification, the inversion a batch shares, the work left in flight when a collision lands. A number that prices one phase is a stage diagnostic, never a speed. Every change is classified by what moved: an *advance* lowers the ratio to the floor; *engineering* lowers the cost with the ratio to the floor flat; *relabelling* lowers a headline while the total rises; *accounting* changes a number without changing the algorithm. All four are committed, and every row in this paper is one of the last three. The same rule, applied across Pollard rho, baby-step giant-step, kangaroo and prime-field index calculus, is the subject of a companion draft in the sibling library [1]; this paper does not depend on it. A rate quoted without its paired control, its hardware and its geometry is the unit mistake this paper tries hardest not to make, and Sections 5.3 and 9.2 each record one instance where an earlier draft of the repository made it.

Availability. Every number in this paper carries a footnote naming the file it came from. The campaign lives in a public study repository, `crypto` (`https://github.com/aburan28/crypto`), whose notes are cited by path relative to its root. A public sibling library, `cryptanalysis` (`https://github.com/aburan28/cryptanalysis`), holds an independent C golden model of the same field, a digit-serial FPGA core with no vendor timing, an import of the 20 B/s GPU client with its own host re-walk, and the boundary ledger for index calculus; its paths are prefixed `cryptanalysis:`. Two evidence records cited in Section 10 live in a third public repository, the research ledger (`https://github.com/aburan28/crypto-autoresearcher`), prefixed `autoresearcher:`. The campaign's live aggregates are published at `https://aburan28.github.io/crypto/status/`; the snapshot this paper reports is committed beside the paper source.

What is not claimed. No logarithm has been recovered on the challenge curve and none is expected from the corpus so far; collection is paused and the paper says so. No rate in this paper is quoted for hardware that was not measured; the L40S collection rate, the RTX PRO 4500 collection rate under the bridge map and the G7e instance family are unmeasured and are said to be. Projections (the 7.2–7.4 B/s fused build on the 4500, the table walk's 0.81–0.86 cost per solve, the 136- and 144-engine F2 images) are labelled as projections where they appear.

2 The curve, the field and the golden model

ECC2K-130 is the Certicom challenge [11]

$$E:\ y^2 + xy = x^3 + 1 \qquad \text{over } \mathbb{F}_{2^{131}},$$

a Koblitz curve [23] defined over $\mathbb{F}_2$, with group order $4 \cdot \ell$,

$$\ell = 680564733841876926932320129493409985129,$$

a 129-bit prime.[1] The challenge asks for k with $Q = [k]P$ for published P and Q of order ℓ. It has been open since 1997; the 2009 consortium estimate of the work is $2^{60.9}$ iterations of a Frobenius-equivariant walk [3], and Section 3.4 measures this paper's walk at $2^{60.91}$–$2^{60.92}$ on completed trails.

The curve is identified rather than trusted. $E(\mathbb{F}_2)$ has four points, so the Frobenius trace is $t = -1$ and the Koblitz recursion $s_m = t\, s_{m-1} - 2s_{m-2}$ gives $\#E(\mathbb{F}_{2^{131}}) = 2^{131} + 1 - s_{131} = 4\ell$. The sibling library's test suite checks the two facts that make this *this* curve, on random points multiplied by the cofactor: $\sigma^2 + \sigma + 2 = 0$, where $\sigma(x, y) = (x^2, y^2)$ is the Frobenius, and

[1] `ecc2k130/README.md, "The problems"`.

$\ell \cdot R = \mathcal{O}$.[2][3] The published challenge point converts out of the normal basis back to Certicom's polynomial-basis hex, `05 1C99BFA6 F18DE467 C80C23B9 8C7994AA`.[4]

2.1 A type-II optimal normal basis

$2 \cdot 131 + 1 = 263$ is prime and 2 has order 131 modulo 263, so $\mathbb{F}_{2^{131}}$ has a type-II optimal normal basis. With ζ a primitive 263rd root of unity the basis is $\gamma_i = \zeta^i + \zeta^{-i}$, $i = 1, \ldots, 131$, and an element is a 131-bit vector. Every design in this paper, on every platform, is built on three consequences.[5]

fact	consequence
$\gamma_i\gamma_j = \gamma_{i+j} + \gamma_{i-j}$ (indices folded by $\gamma_{263-k} = \gamma_k$, $\gamma_0 = 0$)	a product is a cyclic convolution of two symmetric 263-bit vectors, or, after a Bernstein–Lange change of basis, a polynomial product with no reduction step
$\gamma_i^2 = \gamma_{2i \bmod 263}$	squaring is the index permutation $i \mapsto \text{fold}(2i)$: free when loops are unrolled, wiring on an FPGA, so every Frobenius power σ^k costs nothing
the Hamming weight of the vector is invariant under σ and under negation $-(x, y) = (x, x + y)$	the iteration's branch selector and the distinguished-point predicate are functions on $\langle\sigma, -1\rangle$-classes that read the vector the hardware already holds

2.2 The golden model

None of this is taken on trust. The code generator models the field as the symmetric vectors over $\mathbb{Z}/263\mathbb{Z}$ modulo the all-ones vector, with multiplication as cyclic convolution: a complete and obviously correct model of $\mathbb{F}_{2^{131}}$, and the one every generated routine is checked against on 64 random inputs before it is written out.[6][7] The bitsliced multiply, square, inverse, σ^7 and Hamming weight are differentially tested against a reference implementation that shares no code with them; the iteration function is checked to commute with Frobenius and with negation; the collision solver recovers a planted logarithm; the distinguished-point comparison is checked exhaustively over all weights and cutoffs. `make test` runs 52 such checks across GF(2^{23}), GF(2^{41}), GF(2^{83}) and GF(2^{131}).[8]

The same model is the oracle for every other platform. The VHDL engine's reference script imports it directly and proves the basis-change maps, the multiplier route, σ^j, the inversion chain and a full step before emitting test vectors (Section 8).[9] The sibling library's independent C model of the same field and curve (`cryptanalysis:fpga/model/ecc2k130.c`) was built separately and re-walks every GPU report that library imports; agreement between the two models on the same records is equality of 131-bit vectors in the same permuted basis, not equivalence up to a basis change [1].

Longer runs check the reporting path rather than the algebra. On GF(2^{83}), 12.6 M iterations produced 172 distinguished points, within 30% of the predicted rate, and 60 of them were recomputed from their seeds by the reference with no mismatch, including walks that had been restarted. On the challenge curve itself, with the cutoff loosened from 34 to 50 so that points

[2] `cryptanalysis:fpga/README.md`.
[3] `cryptanalysis:fpga/model/ecc2k130_test.c`.
[4] `ecc2k130/README.md, "Validation"`.
[5] `ecc2k130/README.md, "Field representation"`.
[6] `ecc2k130/codegen/field.py`.
[7] `ecc2k130/codegen/gen.py`.
[8] `ecc2k130/README.md, "Validation"`.
[9] `hdl/ecc2k130/ecc2k_ref.py`.

appear, 3.1 M iterations produced 12,368 reports, 25 of which were recomputed and matched exactly.[10]

2.3 ECC2K-95 as the end-to-end control

ECC2K-95 is the retired Certicom challenge on the same curve equation over $\mathbb{F}_{2^{97}} = \mathbb{F}_2[t]/(t^{97} + t^6 + 1)$, group order 4ℓ with $\ell = 39614081257132074233778707191$ a 95-bit prime, solved by Harley and collaborators in 1998 [19]. $2 \cdot 97 + 1 = 195$ is composite, so there is no type-II optimal normal basis; the client runs a polynomial-basis backend with the orbit-invariant weight taken through one generated linear map into a normal basis, which is the arrangement Harley's own client used.[11]

The parameters were recovered from Harley's surviving client source and the generator refuses to emit the curve unless both points lie on it, both have order ℓ, and the published answer

$$k = 37837308472231540269443981458$$

satisfies $[k]P = Q$. Expected work is about 2^{44} iterations; Harley's group measured $2.16 \cdot 10^{13}$ against the $1.79 \cdot 10^{13}$ this walk predicts from $\sqrt{\pi\ell/(4m)}$.[12] Per iteration the polynomial-basis field is about 1.9× cheaper than GF(2^{131}) in the normal basis (about 32,000 against 60,000 bitsliced instructions), so ECC2K-95 is a few GPU-hours where ECC2K-130 is GPU-centuries: a real end-to-end test of the whole pipeline at a scale that completes. [13] The repository's separate ECC2K-95 CUDA kernel has run only through CPU emulation, where it recovers 21-bit and 40-bit toy logarithms at 0.94× and 1.32× its own prediction, and is not a throughput claim.[14]

3 The walk

Pollard's rho [26] in the parallel collision-search form of van Oorschot and Wiener [29] walks many pseudo-random trails through the group, reports the distinguished points it passes, and solves the logarithm when two trails meet. On a Koblitz curve the Frobenius σ and negation generate a group of automorphisms of order $2m = 262$ that act on $\langle P \rangle$, and a walk defined on the orbits of that group needs $\sqrt{262} \approx 16.2$ times fewer steps [13, 17, 30]. The expected number of classes visited before a collision on $N = (\ell - 1)/262$ classes is $\sqrt{\pi N/2}$, which is $2^{60.809}$ here.[15] Every walk in this paper is a map on those classes; what differs between them is how the class action is respected and how much each step costs.

3.1 The σ-walk

The campaign walks the map of Bailey et al. [3]:

$$\begin{aligned} j &= 3 + ((\mathrm{HW}(x_R) \gg 1) \bmod 8) \in [3, 10], \\ R' &= \sigma^j(R) + R, \\ \text{distinguished} &\iff \mathrm{HW}(x_R) \le w, \end{aligned}$$

with $w = 34$ in the client's default and $w = 32$ in the live campaign. Because the weight is a class invariant, $f(\sigma R) = \sigma f(R)$ and $f(-R) = -f(R)$, so the map descends to classes without any canonical representative being computed inside the loop. The division by two in the selector

[10] `ecc2k130/README.md, "Validation".`
[11] `ecc2k130/README.md, "ECC2K-130 and ECC2K-95".`
[12] `ecc2k130/README.md, "The problems".`
[13] `ecc2k130/README.md, "ECC2K-95".`
[14] `gpu/ecc2k/README.md.`
[15] `ecc2k130/WALK-CONSTANT.md, Âğ1.`

is load-bearing: x-coordinates of subgroup points have even weight, and a selector on the raw weight would use only four of the eight branches.[16]

One step is one affine addition, with the inversion shared across a batch (Section 4); in the normal basis σ^j is an index permutation and costs nothing. As a scalar the step is $R \mapsto [1+s^j]R$ where s is the root of $s^2 + s + 2 = 0$ that σ realises on $\langle P \rangle$. The σ^j selection is branch-free: three conditional squarings, each one `LOP3` per word, and j is never materialised.[17]

No coefficients are tracked. A walk is identified by its seed, and a report is the pair (seed, canonical endpoint), 32 bytes. The start point is $R_0 = Q + \sum_i c_i\, \sigma^i(P)$ with c a 128-bit string from a PRF of the seed, computed with the same arithmetic and merged into finished lanes under a mask; tracking a linear combination in the loop would cost a 129-bit modular multiplication per step, which this walk cannot afford. Collision resolution re-walks both trails counting how often each branch $1+s^j$ was applied, which gives endpoint $= [\mu](\alpha_0 P + Q)$ with $\mu = \prod_j (1+s^j)^{n_j}$, matches the two endpoints up to Frobenius and negation, and solves for k modulo ℓ.[18] The σ-walk has no fruitless cycles: no product $\prod_j (1+s^j)^{n_j}$ equals $\pm s^a$, so a trail cannot return to its own class.[19]

The client can also carry the eight branch counters as it goes (`WITNESS=1`). That is eight counters and not a coefficient, because the product μ commutes and nothing is multiplied per step. It turns a report into something a third party checks with one scalar multiplication $[\mu]R_0$, 196 point operations, instead of re-walking the $2^{25.27}$ steps a weight-34 report represents, an asymmetry of about $2^{17.7}$. On the bitsliced CPU walk it costs 6.0% measured, the median of three paired rounds (40.026 to 37.762 M it/s, a 5.7% drop in rate, against 33.8% predicted; an earlier revision measured 4.2% on a pre-merge tree). On the packed GPU walk the rate fell 34.7% (14.003 to 9.150 B/s, stated in the note; only the job script is frozen under `benchmarks/witness-cost/`), far above the 2% the note had set as its abandonment threshold for a 16-bit counter layout that was never built, so the campaign builds pin `WITNESS=0` and the record stays 32 bytes.[20][21] A later fused kernel narrows the packed cost (9.347867 to 11.800903 B/s in the witness-carrying mode) without closing it.[22]

3.2 The r-adding table walk

The σ-walk pays for σ^j in the basis conversions around it on a GPU (Section 4.2). An r-adding walk on the same classes removes that cost:

$$
\begin{aligned}
h(R) &= (\mathrm{HW}(x)/2) \bmod 8, \\
k(R) &= \Big(\textstyle\sum_e L(e)\, x_e\Big) \cdot \mathrm{HW}(x)^{-1} \bmod 131, \\
\varepsilon(R) &= \text{bit } p(R) \text{ of } y, \\
R' &= R + (-1)^{\varepsilon}\, \sigma^k(T_h),
\end{aligned}
$$

with $T_h = [a_h]P + [b_h]Q$ and a 131×8-point table in shared memory (37.7 KB, 48.5 KB per block with the selection tables). k shifts by one under σ and ε flips under negation, so the walk descends to classes; the step is one affine addition with a table operand whose Frobenius power is a lookup rather than a conversion.[23]

Unlike the σ-walk, an adding walk on classes has fruitless cycles [6]. Two classes occur here: four-step patterns in which the table points' Frobenius powers satisfy a relation from

[16] `gpu/ecc2k/README.md`.
[17] `ecc2k130/README.md, "Iteration function"`.
[18] `ecc2k130/README.md, "Distinguished points and restarts"`.
[19] `ecc2k130/WALK-CONSTANT.md, Âğ5`.
[20] `ecc2k130/CAIRN-WITNESS.md`.
[21] `ecc2k130/aws/build.sh`.
[22] `ecc2k130/benchmarks/sigma-fused/RESULTS.md`.
[23] `ecc2k130/ITERATION-FUNCTION.md, Âğ4 and Âğ6.1`.

Table 1: Walk constant c on toy Koblitz curves, ECC2K-130 branch distribution ($H = 8$). Floor $c = 1$. The adding-walk control with 256 branches measures 1.0002 ± 0.0012 at $n = 23$. The injective-branch model $1/\sqrt{1 - \sum p^2}$ predicts 1.070 for the σ-walk at $n = 131$.

n	measured on	table walk	σ-walk
23	host emulation	1.0037 ± 0.0012	1.0995 ± 0.0013
23	device	1.0076 ± 0.0037	1.0948 ± 0.0040
31	host emulation	1.0035	1.0964
37	host emulation ($W = 16$)	1.0036	1.0949
41	host emulation	1.0125 ± 0.0059	1.0910 ± 0.0064
41	device	1.0051 ± 0.0299	1.0997 ± 0.0336
59	host emulation	0.9949	1.0817
pooled / extrapolated to $n = 131$		1.0035 ± 0.0006	[1.070, 1.082]

$\tau^2 + \tau + 2 = 0$, and six-step pairwise patterns. The first cycle rule let 24 four-step and 4 six-step patterns through; the count observed against the count predicted was 1,145 against 1,312 (0.87 ± 0.03) for the τ-relation class and 126 against 166 for the pairwise class.[24] At the campaign's trail length of $2^{28.41}$ steps a model built on the measured rates predicts that those cycles trap 53.7% of trails, a loss of 7.1–8.6× with the 2^{30} iteration guard.[25] A revised rule (v2) brings residual trapping to $2.0 \cdot 10^{-5}$ of trails at a 1.0048× iteration cost from merge parting; a third revision (v3, Section 3.5) replaces history-dependent exits with a raw-cycle anchor.[26]

3.3 The x-only doubling-and-bridge map

On the power-capped RTX PRO 4500 a third map pays. The common transition is $R' = R + \sigma(R)$, which as a scalar is $[1 + s]$; on this curve $x(R + \sigma R) = x + 1/x$, the square root of $x([2]R)$, so the walk keeps an x-only state. That scalar alone generates a subgroup of index 12, so a sparse bridge $R' = R + \sigma^3(R)$ (index 109 on its own) is taken when $\mathrm{HW}(x) \bmod 72 = 14$; together the two generate the full group, which the earlier [2]/[3] map had failed to do by an index of two ("the scalar subgroup defect").[27] A frozen study of 2,000 trials on GF(2^{23}), where the modulus-72 bridge predicate selects exactly what the earlier modulus-32 one did (weight 14 only), gives mean charged work 1.006696× that of the then-selected map, so the map's throughput gain (Section 5.5) is not paid back in extra iterations. The map passes an independent 2,048-point affine checkpoint.[28]

3.4 The walk constant, measured

Throughput is half of a rate. The other half is the number of iterations a walk needs, which for a non-random map differs from the random-mapping $\sqrt{\pi N/2}$ by a constant c. The repository measures c as iterations divided by a random mapping's iterations on the class set, on toy Koblitz curves of the same family, with the device walk and a host emulation measured separately and an r-adding control with 256 branches alongside.[29]

The σ-walk is a one-to-one map on each branch and pays the injective-branch penalty $1/\sqrt{1 - \sum_j p_j^2}$, which with the weight distribution at $n = 131$ ($\sum p_j^2 = 0.127$) is 1.070; the measurements in Table 1 drift from 1.0995 at $n = 23$ toward it, and the extrapolation to

[24] `ecc2k130/WALK-CONSTANT.md, Ãğ5.`
[25] `ecc2k130/WALK-CONSTANT.md, Ãğ6.`
[26] `ecc2k130/WALK-CONSTANT.md, Ãğ11.1.`
[27] `ecc2k130/README.md, "ECC2K-130 and ECC2K-95".`
[28] `ecc2k130/benchmarks/g7-12b-xonly-doubling-bridge/README.md.`
[29] `ecc2k130/WALK-CONSTANT.md, Ãğ4.`

$n = 131$ is $[1.070, 1.082]$. The table walk is not injective per branch and sits at 1.0035 ± 0.0006 pooled. The ratio c_t/c_σ is 0.928–0.938. Expected work for the campaign's σ-walk is therefore

$$2^{60.809} \times [1.070,\ 1.082] = 2^{60.91}\text{–}2^{60.92}$$

iterations, with a further 1.185× (that is, $2^{61.15}$–$2^{61.17}$) under the original 2^{30} restart guard: a guard at three trail lengths cuts 4.9% of trails, the longest ones, and so discards 15.6% of steps (a model on the measured rates). Bailey et al.'s $2^{60.9}$ stands.[30] The guard was raised to 2^{32}, twelve trail lengths at weight 32, which discards 0.007% of steps (1.00007×).[31]

On GF(2^{41}) with 128 walks, cutoff 13 and 64 planted logarithms, the σ-walk needed 164,200± 9,900 iterations per solve and the table walk 157,800 ± 7,500, ratio 0.96 ± 0.07, against a bare prediction of 102,600 with about 1.5× harness overhead from the distinguished-point tail.[32]

3.5 Cost per solve, not rate

Putting rate and constant together is where the table walk's story changes. Its throughput advantage of Section 5.4 is 1.09–1.15×; its walk constant is 0.93×; but its as-built cycle rule is modelled to trap half of all campaign-length trails, so the cost per solve of the as-built kernel was 4.85–6.26× the σ-walk's. With rule v2 the projection is 0.81–0.86×, and the switch deadline (the fraction of the campaign after which switching no longer pays) is 15–22%. That projection has not been tested: the paired rate of the v2 kernel that would test it is declared in the note and has not been run.[33] The campaign therefore still walks the σ-map, and the 20 B/s table-walk rows of Section 5.4 are rates, not costs per solve.

Rule v2 itself passed 19,200 refused τ-cycles in a dedicated test, a 4,096-point host probe with no mismatch and planted logarithms 8/8, and its scatter check against the predicted cycle counts gives $\chi^2 = 8.79$ on 15 degrees of freedom (the v1 rule: 23.30).[34] Rule v3 then replaced the history-dependent exits with a raw-cycle probe of at most eight steps and a least-key anchor exit, checked by regression tests on a raw two-cycle found on the challenge curve and on synthetic entry-phase cycles, with 300 of 300 device reports re-walked and identical corpora across arms; v2's walk-constant and throughput figures do not transfer to it.[35] V3 requires a new table corpus; the campaign's σ-walk corpus is unaffected.

4 Field arithmetic: two representations, one model

A step is dominated by field multiplication: two products for the addition and, with W walks sharing one inversion by Montgomery's trick [24], $3(W-1)/W$ more for the product tree, plus $1/W$ of an Itoh–Tsujii inversion [21] that costs eight multiplications in a basis where squaring is free (the client's addition chain is $1, 2, 4, 8, 16, 32, 64, 65, 130$, the FPGA's $1, 2, 4, \ldots, 128, 130$).[36][37] Going from $W = 4$ to $W = 32$ is worth 60% on the CPU client because it drives the amortised inversion from 25% to 3% of the multiplication budget.[38] The representation of a multiplication is therefore the design decision, and the repository carries two.

[30] `ecc2k130/WALK-CONSTANT.md, Âğ6.`
[31] `ecc2k130/benchmarks/max-iters/README.md.`
[32] `ecc2k130/ITERATION-FUNCTION.md, Âğ6.2.`
[33] `ecc2k130/WALK-CONSTANT.md, Âğ11.`
[34] `ecc2k130/WALK-CONSTANT.md, Âğ11.1 and Âğ11.6.1.`
[35] `ecc2k130/CYCLE-ESCAPE-V3.md.`
[36] `ecc2k130/include/packed131.h.`
[37] `hdl/ecc2k130/README.md.`
[38] `ecc2k130/README.md, "Throughput".`

4.1 Bitsliced: 32 walks in the bits of a word

The original client is bitsliced in the sense of Biham [8]: one word operation advances W walks at once, one per bit lane, and a field operation is a straight-line program of word-wide Boolean instructions generated from the field model. The device uses 32-bit words; the host picks the widest it has (AVX-512 gives 512 lanes, AVX2 256, NEON 128), and the arithmetic source is identical at every width.[39]

Multiplication follows Bernstein and Lange [5]: convert both operands from the normal basis to the optimal polynomial basis $\{1, c, c^2, \ldots\}$ with $c = \gamma_1$, multiply as polynomials, and convert the 261-coefficient product back. Both conversions are generated from the substitution $c = z + 1/z$, which splits by parity as $H(c) = H_{\text{even}}(c^2) + c\,H_{\text{odd}}(c^2)$; since squaring is an index permutation this gives an $O(K \log K)$ conversion. The generated `multPrep` is 331 instructions and the double-size inverse conversion 903, against 325 and 909 for the hand-designed chains in the literature.[40]

Two findings from this backend change the standard accounting. First, three-input logic fusion (`LOP3` on NVIDIA, `vpternlogd` on AVX-512, `BSL/EOR3` on Arm with `FEAT_SHA3`) is worth 1.65×: a 131-bit multiplication is 14,554 two-input bit operations but 8,859 instructions once $x \oplus y \oplus z$ and $x \oplus (y \wedge z)$ are single instructions. Because the fused form charges nothing for the XOR that accompanies an AND, schoolbook leaves beat Karatsuba leaves [22]: full recursion costs 18,819 bit operations and 14,770 instructions, while stopping at a 17-bit schoolbook leaf costs more bit operations and fewer instructions. A search over balanced and unbalanced splits found nothing below 8,859; Toom-3 over $\mathbb{F}_2$, the source of Bernstein's 11,961-bit-operation chain [4], is not implemented.[41] Second, code size rather than arithmetic is the wall: inlining everything produced 285,096 PTX instructions for one kernel and `ptxas` had not finished register allocation after five minutes; marking the multiplication, inversion, conversions, leaf, weight tree and σ^j as real subroutines brings it to 36,553 lines and a 2.4 s assembly.[42]

The Hamming-weight tree is 268 instructions against 654 bit operations, because a full adder is two `LOP3`s. The whole bitsliced iteration is about 60,000 instructions per 32 walks on GF(2^{131}), a static count; on the GPU that is the 0.85 B/s control of Section 5.3, and the instruction-count ceiling of the later packed kernel on the RTX PRO 6000 is 17.6 B/s (27.1 T lane-operations/s over about 1,540 lane-operations per update, a different count).[43]

4.2 Packed: one walk per thread, carry-less products

The packed backend holds one walk's coordinates in five 32-bit words per thread and multiplies with carry-less instructions.[44] Coordinates live in $\mathbb{F}_2[c]/(m)$ with $c = \gamma_1$, so a 131-bit product is a Karatsuba over carry-less multiplies, six `CLMAD`, followed by a reduction;[45] the normal basis, where the weight, the branch selection and the distinguished-point test live, is reached through the same Bernstein–Lange conversions, realised as shift-and-mask networks and, for the Frobenius powers, as Beneš permutation networks.[46] Before CUDA 13.3 the carry-less product was built from integer multiplies; the native `clmad.lo/hi` instructions on Blackwell gave +22.4% in Section 5.3 and are the reason the campaign preset requires that compiler.[47]

The accounting unit changes with the representation. In the packed kernel the shipping walk

[39] `ecc2k130/README.md, "Word width".`
[40] `ecc2k130/README.md, "Findings".`
[41] `ecc2k130/README.md, "What is not done".`
[42] `ecc2k130/README.md, "Findings".`
[43] `ecc2k130/THROUGHPUT-CEILING.md.`
[44] `ecc2k130/PACKED.md.`
[45] `ecc2k130/ITERATION-FUNCTION.md, Âğ1.`
[46] `ecc2k130/FROBENIUS-NETWORK.md.`
[47] `ecc2k130/NATIVE-CARRYLESS.md.`

issues 4.81 products per update at batch 16 and about 2,324 static ALU slots;[48] the step count that matters is the `CLMAD` count, and Section 5.1 prices it. The reduction `reducePolynomial131` alone is 440.6 lane-instructions per update, 29% of the integer work of the 20 B/s build.[49]

4.3 Both at once, and bit-for-bit

The two backends, the CPU client at every word width, the VHDL engine and the sibling library's imported kernel are kept bit-for-bit compatible: a distinguished-point record written by any of them re-walks under any other, and every GPU report is re-walked on the host reference before a rate is recorded. The host check re-slices packed words into the same permuted type-II basis and compares vectors, not classes.[50] The exceptional case $\sigma^j(R) = \pm R$, which makes the addition's denominator zero, has probability about 2^{-131} per step. The σ-walk client and the VHDL engine do not special-case it: a zero denominator inverts to zero and the batch's points become garbage, which the next distinguished-point check or the guard discards. The sibling C model and the table walk abort the step instead.[51][52]

5 The GPU client

The GPU question is a throughput question at a known exponent. The unit throughout is billions of complete scalar updates per second on one card, written B/s; a scalar update is one complete iteration of one walk, including its share of the batched inversion, the distinguished-point test and the restart path. The class of every row in this section is engineering: the walk is the same, the collision structure is the same, and the ratio to the generic-group floor does not move. What moves is the ratio to a second boundary, the carry-less unit of the GPU, derived in Section 5.1 before the kernel was tuned toward it.

5.1 Two boundaries, written down first

Any walk that keeps the rho collision structure performs one affine addition per iteration. With Montgomery's batched inversion [24] across W walks, an affine addition costs a fixed number of field products plus $1/W$ of an inversion. In the packed representation a 131×131 product is a two-way Karatsuba over 64×64 carry-less multiplies, six `CLMAD` instructions. At batch $W = 16$ the shipping kernel issues 77 products per 16 slots, 4.81 per update. Table 2 prices one update on the RTX PRO 6000 (188 SMs).[53]

The two pipes are independent and each writes its own floor. On the carry-less unit, 28.9 `CLMAD` at 1.65 lanes per SM-clock is 17.5 SM-clocks per update, a ceiling of 24.7–26.1 B/s with every other cost free. On the logic pipe, about 1,090 ALU slots already exceed the 959–1,013 that 28 B/s allows. At 90% of either pipe the ceiling is 22.2–23.4 B/s. The question the note was written to answer, whether a different iteration function could take one card from the audited 14.64 to 28 B/s, is therefore answered before any kernel is built: 28 B/s is below the floor of any walk that performs one affine addition per iteration.

The carry-less rate itself was revised upward as the kernels got closer to it. A kernel-mix measurement on the 6000 at 1.62 lane-`CLMAD`s per SM-clock priced the 20 B/s table-walk kernel at 33.1 `CLMAD` per update (28.9 products, 3.0 inversion, 1.25 normal-basis squarings) and wrote a floor of 22.3 B/s.[54] A screened (not yet verified) three-limb Karatsuba arm then implied at

[48] `ecc2k130/ITERATION-FUNCTION.md, Âğ1 and Âğ3.`
[49] `ecc2k130/ROOFLINE.md.`
[50] `ecc2k130/README.md, "Validation".`
[51] `ecc2k130/include/ref.h.`
[52] `hdl/ecc2k130/README.md.`
[53] `ecc2k130/ITERATION-FUNCTION.md, Âğ1.`
[54] `ecc2k130/TWO-CHAINS.md, Âğ1.`

Table 2: Arithmetic floor of one affine-addition update on the RTX PRO 6000, and the budgets three throughput targets allow at 100% pipe utilisation. The carry-less unit issues 1.65 lane operations per SM-clock (1/37.7 of the logic pipe's 62.1, measured on an RTX PRO 4500 with the same SM and later confirmed on the 6000 at 1/38.0); sustained clock 2.30 GHz, maximum 2.43 GHz. Both floors sit above the 28 B/s budget.

component	`CLMAD`/upd.	ALU slots/upd.
4.81 products (6 `CLMAD` each)	28.9	≈ 830
squarings	5.0	≈ 80
batched inversion (/16)	4.6	≈ 270
floor	38.5 (28.9, squarings on ALU)	1,090–1,180
budget at 14.64 B/s	48.7–51.4	1,834–1,938
budget at 20 B/s	35.6–37.6	1,343–1,418
budget at 28 B/s	25.4–26.9	959–1,013

Table 3: Roofline of the 20.078 B/s table-walk build on the RTX PRO 6000: 188 SMs at 2.415 GHz, 22.61 SM-clocks per update. Work per update is measured lane-instructions; the ceiling is the rate at which that unit alone would saturate.

unit	peak lanes/SM-clk	work/update	ceiling (B/s)	busy
ALU (logic)	64	1,319.6	22.0	91.2%
`CLMAD`	2.00	33.1	27.4	73.2%
issue	128	1,698.9	34.2	58.7%
load/store	16	80.0	90.8	22.1%
FMA	64	188.6	154	13.0%

least 1.654 lane-`CLMAD`s per SM-clock, refuting that figure, and the roofline of the 20.078 B/s build (Table 3) puts the carry-less speed of light at 27.4 B/s with the unit 2.00 lanes wide, while the logic pipe at 1,319.6 lane-instructions per update is 91.2% busy and binds first.[55] Thirty B/s would need at most 30.3 `CLMAD` and 968 ALU slots per update, a 27% cut in integer work the note calls "a two-card number". The 28 B/s verdict survives the revision because it was never only a carry-less verdict.

The same note measures the carry-less unit across dies: 37.8 logic slots per `CLMAD.lo` on the 6000, 3.78 on the B200 and 2.00 on the H100, so on the data-centre parts the logic pipe binds by a wide margin and on the B200 the carry-less floor alone would be 255 B/s.[56]

The static accounting behind Table 2 was itself corrected once: an earlier SASS counter reported about 1,570 ALU slots and 29 `CLMAD` per update; the replacement counts 2,271 ALU and 45 `CLMAD` static, 2,324 once `POPC` and `FLO` are priced at their measured quarter rate.[57] The floor argument does not depend on which count is used; the budgets are below both.

5.2 Acceptance rules

With two exceptions named in Section 5.7, every rung in this section is a paired comparison on one GPU allocation, candidate and control interleaved, three to six repetitions each, with the candidate's ratio to the control reported per pair. A change is adopted only when it clears a rule declared before the run: at least 1% on both the benchmark and the DP34 collection workload with every pair favouring the candidate, or, for the later table-walk work, a geometric-

[55] `ecc2k130/ROOFLINE.md`.
[56] `ecc2k130/TWO-CHAINS.md, Âğ6.1-6.2`.
[57] `ecc2k130/ITERATION-FUNCTION.md, Âğ3`.

Table 4: Packed-backend rungs on one RTX PRO 6000. Each gain is the note's own paired comparison against the adjacent control, not a ratio of non-adjacent headlines. Blocks/SM is the resident-block bound; "conf." marks a comparison that also changed occupancy, compiler or walk population (the packed row's control ran 49 M walks once and its candidate 1.5 M walks three times). Class: engineering.

rung	control	candidate	gain
bitsliced control	—	0.852294	baseline
packed normal basis, five words (conf.)	0.852294	2.957961	3.47×
single polynomial product, 2 blocks/SM	2.956981	3.366572	+13.85%
denominator cache, 4 blocks/SM	3.506509	4.024103	+14.76%
same, 6 blocks/SM (conf.)	3.506509	4.094367	+16.76%
Frobenius permutation networks	4.115575	5.811421	+41.21%
polynomial-basis batch chain (conf.)	5.832342	6.054107	+3.80%
paired products (conf.)	6.123614	6.215430	+1.50%
polynomial-basis state	6.301859	6.547850	+3.90%
direct reduction	6.826154	6.906059	+1.17%
generated straight-line product (conf.)	6.860749	6.924275	+0.93%
256-worker tiles	6.852888	6.994745	+2.07%
native `clmad` (CUDA 13.3)	7.110440	8.703518	+22.40%
batch 16, 385,024 workers	8.673447	13.206088	+52.26%
weighted prefixes, paired Frobenius	13.110335	13.323276	+1.62%
compact 17-byte state	13.548376	14.403112	+6.31%
shared-memory mask table	14.296584	14.411102	+0.80%
public audit of that build	—	14.637530	DP34 14.106673

mean ratio at or above a stated threshold with every pair above a stated minimum.[58][59] Every adopted kernel re-walks every device report on the host reference (300 of 300 in the later notes) and recovers planted logarithms on GF(2^{41}). A candidate that fails the rule is kept in the tree behind a flag and recorded with its ratio; the record of declined work is in Section 5.7.

5.3 The packed-backend ledger: 0.85 to 14.64 B/s

The original client was bitsliced: 32 walks in the bit lanes of a word, every field operation a straight-line program of `LOP3` instructions (Section 4.1). The audit of the packed replacement measured that control at 0.852294 B/s on an RTX PRO 6000.[60] Table 4 is the lineage from there to the audited 14.637530 B/s (14.106673 B/s while collecting at weight 34), each rung the note's own paired comparison. The rungs are not a single chain of equal occupancy and equal compiler; where a note changed more than one thing the table says so.

Sources, one per rung: packed basis [61]; single product [62]; denominator cache [63]; Frobenius networks [64]; polynomial chain [65]; paired products [66]; polynomial state [67]; direct reduction [68];

[58] `ecc2k130/NATIVE-CANDIDATES.md`.
[59] `ecc2k130/SIGMA-TABLE.md`.
[60] `ecc2k130/PACKED.md`.
[61] `ecc2k130/PACKED.md`.
[62] `ecc2k130/SINGLE-PRODUCT.md`.
[63] `ecc2k130/DENOMINATOR-CACHE.md`.
[64] `ecc2k130/FROBENIUS-NETWORK.md`.
[65] `ecc2k130/POLYNOMIAL-CHAIN.md`.
[66] `ecc2k130/PAIRED-PRODUCTS.md`.
[67] `ecc2k130/POLYNOMIAL-STATE.md`.
[68] `ecc2k130/DIRECT-REDUCTION.md`.

generated product [69]; tiles [70]; native carry-less [71]; batch tuning [72]; weighted prefixes [73]; compact state [74]; shared masks [75].

Three rungs carry most of the factor and each is a representation change rather than a tuning. The Frobenius permutation networks (+41%) replace the per-step basis conversions around σ^j with Beneš networks over the permuted normal basis, checked on 3,120 GPU vectors and on 256 basis vectors under all 131 powers. Native carry-less multiplication (+22%) replaces an emulated 32×32 carry-less product with the `clmad.lo/hi` instructions that CUDA 13.3 exposes on Blackwell; the same change later let the batch fall from 32 to 16 walks per inversion with 385,024 workers resident (+52%), at the same two-block occupancy and the same 122 registers per thread; the note suggests, but did not measure, a cache-capacity explanation (hot state falls from 245 to 122 MiB).[76] The compounded remainder between the Frobenius-network confirmation (5.81 B/s) and the shipping walk (14.64 B/s) is those two changes plus layout work; the table gives each its own pair, and no single "remaining factor" is quoted as a measurement.

The σ-walk itself did not stop at 14.64. A fused single-pass variant measured 15.436677 against 15.015004 B/s (1.028×, five pairs), and a σ step through nibble tables in shared memory (68,352 bytes, 512×1 blocks) measured 15.884369 against 15.513653 B/s, paired median 1.023930 with every pair at or above 1.0238 and an A/A drift of 0.099%, clearing a pre-registered gate of geometric mean ≥ 1.015 and every pair ≥ 1.005.[77][78] Those are the current σ-walk headlines; the campaign configuration of Section 9 was frozen before them and still ships the audited 14.64 build.

5.4 Beyond the shipping walk: the table walk and 20 B/s

The σ-walk spends part of every step on σ^j and the basis conversions around it. A redesigned r-adding table walk (Section 3.2) removes both: the step adds a precomputed table point $\pm\sigma^k(T_h)$ whose Frobenius power is a shared-memory lookup, and the static ALU count falls from 2,324 to 1,831 slots per update. On the same host and automatic geometry it measured 16.560337 B/s against 14.412634 B/s for the shipping walk (+14.9%, six alternating repetitions, every paired ratio between 1.146 and 1.169); in the audited 385,024-worker geometry 16.351936 against 14.975448 (+9.2%). Three hundred of 300 device reports re-walked and 64 of 64 planted logarithms were recovered on GF(2^{41}).[79][80]

The note had set 17.0 B/s as its own target (§4.4) before the build. Both medians missed it, by 2.6% and 3.8%, so by the rule declared there the table walk was declined as the campaign default and left available behind `WALK_TABLE=1`. That is the acceptance rule working in the direction a rule exists for.

The table walk also changes the walk constant, and not in its favour as first built: Section 3.4 measures its cycle trapping at dpWeight 32 and prices the as-built kernel at 4.85–6.26 times the σ-walk's cost per solve, with a revised cycle rule projected, not yet measured, at 0.81–0.86. The throughput rows here are therefore rates, not costs per solve.

Under a different geometry the same walk goes further. A one-block layout (512 threads and one resident block per SM, which gives the block 64 KB of L1 instead of 28 KB), denominators

[69] `ecc2k130/GENERATED-PRODUCT.md`.
[70] `ecc2k130/TILED-STATE.md`.
[71] `ecc2k130/NATIVE-CARRYLESS.md`.
[72] `ecc2k130/BATCH-TUNING.md`.
[73] `ecc2k130/WEIGHTED-PREFIX.md`.
[74] `ecc2k130/COMPACT-STATE.md`.
[75] `ecc2k130/SHARED-SIGMA.md`.
[76] `ecc2k130/BATCH-TUNING.md`.
[77] `ecc2k130/benchmarks/sigma-fused/RESULTS.md`.
[78] `ecc2k130/SIGMA-TABLE.md`.
[79] `ecc2k130/ITERATION-FUNCTION.md, Âğ6.2-6.3`.
[80] `ecc2k130/benchmarks/table-walk/comparison.json`.

rebuilt from the step tag instead of stored, inlined products and a pipelined selection pass took a 17.406 B/s reference to 20.078 B/s (+15.3%, five samples in 20.077–20.080), 0.73 of the 27.4 B/s carry-less speed of light (0.90 of the since-refuted 22.3); the geometry alone was +9.4%, tag denominators +1.5%, inlining +2.3%. Collecting at weight 34 it holds 19.04 B/s against 16.70. Three hundred of 300 reports re-walked and the distinguished-point count matched the reference's 472,776.[81] This is the `gpu-rtx-pro6000-20b` build. A cycle-escape revision of the table walk (rule v3, Section 3.5) then rebuilt the kernel shape; its own ladder from 1.04 to 5.10 B/s is a different walk identity and is not comparable to the 20 B/s rows.[82]

5.5 The second card: RTX PRO 4500

The campaign's second GPU family is the RTX PRO 4500 (82 SMs, 165 W) in the AWS `g7.2xlarge`. Its ledger is separate because the power cap, not the pipe, binds: the shipping walk measures 5.107 B/s here at 1.95 GHz, within about 1% of what the 6000's per-SM-clock rate predicts (Section 9.3), the card's own selected preset before the map change sat near 6.2 B/s, and the larger split-batch candidates, for example, were slower because they raised memory activity and lowered the clock under the cap.[83][84] A pre-declared goal of 12 B/s on this card (Section 9.3) was reached by a map change rather than a kernel change: the complete two-bridge x-only map of Section 3.3 measured 12.499203 B/s median against 11.218437 B/s for its matched modulus-32 control, a paired geometric-mean ratio of 1.121079 with 95% interval [1.105668, 1.136704]; a fresh three-run confirmation gave 12.567029, 12.457221 and 12.366415 B/s, and the production build 12.632704 B/s over 2^{36} updates.[85][86] Its DP34 collection rate on this card is unmeasured and the paper does not supply one.

5.6 Per-GPU survey

The campaign default is the RTX PRO 6000. Modal also rents T4, L4, A10, L40S, A100, H100, H200, B200 and B300. Table 5 is one packed walk, automatic workers, three valid `-bench` repeats, on twelve catalogue SKUs. No row beat the 6000. The walk is integer- and carry-less-bound, not tensor-core-bound, and LLM SKU rankings do not transfer: the B200 is 0.585× the 6000 on this kernel and the H100 half. [87]

On the B200 the table-walk build (15.365 B/s) was later lifted by an automatic sweep over twenty single-knob arms (launch geometry, product form, scheduling, memory residency) to 19.402 B/s (1.2627, every pair in 1.2623–1.2632) with a combination of the fused pass, batch 32, three-limb Karatsuba and 512-thread blocks, adopted as `gpu-b200-19b`. The same sweep on the 6000 found a 640×1 arm at 19.993 against 19.951 with two of five pairs below one, and adopted nothing.[88] The trades that flip sign with the die are the ones that move work from the logic pipe onto the carry-less unit: the top-bit correction on `CLMAD` costs 15% on the 6000 and gains 13.4% on the B200, and stacked with the normal-basis inversion and λ^2 on `CLMAD` it reaches 1.379× (15.372 B/s) there, which is what the per-die `CLMAD` rates of Section 5.1 predict. A two-chain kernel that splits each warp's Montgomery chain in two lost on both dies (0.826× on the B200, 0.696× on the 6000), a register-file effect rather than a unit-rate one.[89]

[81] `ecc2k130/ONE-BLOCK-GEOMETRY.md`.
[82] `ecc2k130/CYCLE-ESCAPE-V3.md`.
[83] `ecc2k130/ITERATION-FUNCTION.md, Âğ6.3`.
[84] `ecc2k130/README.md, "larger split-batch experiment"`.
[85] `ecc2k130/README.md, "ECC2K-130 and ECC2K-95"`.
[86] `ecc2k130/benchmarks/g7-12b-xonly-doubling-bridge/README.md`.
[87] `ecc2k130/benchmarks/modal-gpus/SURVEY.md`.
[88] `ecc2k130/AUTOSWEEP.md`.
[89] `ecc2k130/TWO-CHAINS.md, Âğ5, Âğ6.2 and Âğ6.4`.

Table 5: Modal GPU survey, automatic workers, packed recipe. Unit: billion complete iterations/s. The last column is trillions of iterations per dollar at Modal list price. Class: engineering.

GPU	B/s	/ 6000	it/$ (T)
RTX PRO 6000, 385k workers	15.116	1.000	17.95
RTX PRO 6000, automatic	14.470	0.957	17.19
B200	8.840	0.585	5.09
L40S	8.696	0.575	16.04
H200	7.632	0.505	6.05
B300	7.575	0.501	3.84
H100	7.540	0.499	6.87
A100-80GB	4.632	0.306	6.67
A100-40GB	4.618	0.305	7.92
L4	2.561	0.169	11.54
A10	2.423	0.160	7.92
T4 (no `clmad`)	0.541	0.036	3.30

5.7 What was declined

Table 6 lists every GPU candidate that was built, measured against the rule, and not adopted. They are recorded so that a later thread which rediscovers one finds the measurement rather than the folklore. Each row passed its correctness gates; what failed was the rate.

Sources: stream Karatsuba [90]; add-combine [91]; top-`CLMAD` [92]; batch screen [93]; two chains [94]; λ-projective [95]; the 4500 rows [96], each linking its `benchmarks/g7-*` directory.

Two of the rows deserve a sentence. The λ-projective row was never built: pricing it in the unit of Table 2 gave 60.5 `CLMAD` per update against 38.1 for affine at batch 16, a ceiling of 11.8–12.5 B/s below the 14.41 already measured, so the pre-declared condition (a negative cost difference at $W = 16$) was unsatisfiable and the build was not started. The top-`CLMAD` and stream-Karatsuba rows are the two comparisons in this section that were not interleaved on one allocation (the stream-Karatsuba pair also differs in source digest); both are reported with that caveat and neither would have cleared the rule regardless.

6 CPU clients

The CPU client is the bitsliced backend at the host's word width. It matters for three reasons: it is the verification path for every GPU report, it is the ECC2K-95 end-to-end control, and the campaign's spot fleet included CPU workers whose points are indistinguishable from the GPUs'.

Table 7 is the ledger.[97] Two of its facts generalise. The AVX-512 build spends half its multiply on moves rather than arithmetic (4,069 logic instructions against 4,287 `vmovdqa32`) because the Karatsuba leaf keeps 254 values live against 32 registers; a scheduling pass over the same DAG recovered moves only in the conversion (4,287 to 3,903), missed its declared target, and recorded the leaf-size and compiler levers as measured dead ends.[98] On aarch64 the

[90] `ecc2k130/TUNING.md`.
[91] `ecc2k130/ADD-COMBINE.md`.
[92] `ecc2k130/TOP-CLMAD.md`.
[93] `ecc2k130/COMPACT-STATE.md`.
[94] `ecc2k130/TWO-CHAINS.md`.
[95] `ecc2k130/LAMBDA-PROJECTIVE.md`.
[96] `ecc2k130/README.md, "ECC2K-130 and ECC2K-95"`.
[97] `ecc2k130/README.md, "Throughput"`.
[98] `ecc2k130/HOST-SCHEDULE.md`.

Table 6: GPU candidates measured and declined. Ratio is candidate over the paired control; intervals are 95% paired bootstrap where the note reports one. "4500" rows are on the RTX PRO 4500 against the selected runtime of the day; the others are on the RTX PRO 6000.

candidate	control	candidate	ratio
stream Karatsuba compile option	0.857	0.748	0.873
integer add in place of OR in `clmul32`	6.8615	6.8080	0.992
top three-bit correction moved onto `CLMAD`	15.116	12.859	0.851
batch 12 / 10 against batch 16	14.26	13.75 / 13.70	$<$ screen
batch 18 against batch 16 (larger population)	14.47	14.29	$<$ screen
table walk against pre-declared 17.0 B/s	14.41	16.56	1.149, missed
two Montgomery chains per warp	19.981	13.903	0.696
λ-projective coordinates (priced, not built)	—	ceiling 11.8–12.5	< 14.41
4500: hybrid reduction	6.2459	5.3459	0.852 $[0.845, 0.860]$
4500: half-native product tail	6.1898	6.0303	0.977 $[0.971, 0.982]$
4500: quotient fan-in	6.1739	6.0139	0.977 $[0.972, 0.982]$
4500: register-retained inverse, 3 / 4 blocks	6.1724	6.1487 / 5.6922	0.997 / 0.924
4500: paired inverse leaves, 3 / 4 blocks	6.1643	6.1146 / 5.6501	0.994 / 0.918
4500: logical-pair inversion, 3 / 4 blocks	6.1675	6.1037 / 5.6723	0.960 / 0.921
4500: seven-node inversion, 3 / 4 blocks	6.1826	6.1388 / 5.6904	0.995 / 0.923
4500: seven-node shared-X cache, six slots	6.2624	6.3741	1.010 $[0.991, 1.029]$
4500: split batch 24 / 32	6.2742	6.1218 / 5.9950	0.972 / 0.948
4500: direct polynomial δ-3/4 routing	6.2724	4.3635	0.690 $[0.676, 0.704]$
4500: 65/66-bit, 33-bit products; byte masks (four)	—	—	all < 1

gain splits cleanly: the NEON word with `BSL` alone is 1.76× on the M4 Pro, `EOR3` a further 1.24×, and a core without `FEAT_SHA3` (Graviton2) still takes 1.81× from the wider word and `BSL`. Compiler choice is worth more than either on this code: gcc emits 9,372 instructions for the multiply leaf, 59% of them loads and stores against spilled temporaries, where clang emits 1,902 plus five outlined calls. The SHA3 probe is gated on the CPU reporting the feature, because an ungated probe on a Graviton2 selects a `+sha3` spelling under both compilers and the resulting binary dies with `SIGILL` on its 8,907 `eor3` instructions.[99]

In the fleet a CPU worker at 2.3 B/s (the self-reported median of the twelve CPU slots on 2026-09-17) is about a sixth of an RTX PRO 6000; on Modal a 32-thread (16-core) sidecar beside a GPU adds about 25% to the container's cost for 2.7% more iterations and is off by default.[100][101]

7 The other GPU kernels: prime fields and binary fields beyond ECC2K-130

The ECC2K-130 client is one of several GPU arithmetic libraries in the two repositories. The others target the 256-bit prime field of secp256k1 (a Koblitz curve in the SEC sense, with $j = 0$ and the order-3 endomorphism), the NIST primes, the polynomial-basis field of ECC2K-95, and the linear algebra and polynomial-system solvers that index calculus needs. They share the ECC2K-130 client's discipline: a Python or Rust oracle, a host emulation of the identical device source, bit-identical steppers, planted logarithms, and static instruction accounting from the assembler. What most of them lack is a measurement on a device. This section says which is which before it says anything else, because a static instruction count and a measured rate are

[99] `ecc2k130/README.md, "Word width".`
[100] `ecc2k130/benchmarks/dp-interval/raw/worker-counters-2026-09-17.tsv.`
[101] `ecc2k130/README.md, "Walking on the container's CPUs too".`

Table 7: Bitsliced CPU client, GF(2^{131}), one core, median of five. Speedups are against the g++ 64-lane build on the same host. The 2009 qhasm implementation of Bailey et al. reached 533 cycles per iteration on a Core 2 with 128-bit vectors.

host	word	lanes	M it/s	note
Xeon 2.1 GHz	`uint64_t`	64	4.42	475 cycles/iteration
Xeon 2.1 GHz	AVX2	256	10.7	196 cycles/iteration
Xeon 2.1 GHz	AVX-512	512	12.6	167 cycles/it.; 4 threads 40 M
M4 Pro, g++ 16.1	NEON	128	11.342	2.19× over 64 lanes (5.185)
M4 Pro, Apple clang 17	NEON	128	15.673	3.02× over g++/64
Graviton3 (V1, SHA3), clang	NEON	128	6.214	2.38× over g++/64
Graviton2 (N1, no SHA3), clang	NEON	128	4.220	2.09× over g++/64
Modal container, 4 threads	AVX-512	512	12.4	49.4 M it/s on 4 threads

Table 8: GPU arithmetic in the two repositories by evidence class. "Emulated" means the device source was executed on the host and checked against an oracle; "static" means instruction, register and stack counts from `ptxas` with no execution.

kernel family	field	on a device	otherwise
ECC2K-130 client (Sections 5, 9)	$\mathbb{F}_{2^{131}}$, ONB	measured, audited	emulated, static
F4 Gröbner elimination (sibling)	$\mathbb{F}_2$ Macaulay matrices	measured (RTX 5090, H100)	emulated
Macaulay reduced echelon	$\mathbb{F}_p$, $p = 2083$	one scheduling run (RTX PRO 4500)	emulated, static
secp256k1 rho, baby-step giant-step	$\mathbb{F}_p$, 256-bit	never	emulated, static
secp256k1 kangaroo	$\mathbb{F}_p$, 256-bit	never	emulated, static
P-256 / P-384 arithmetic	$\mathbb{F}_p$, 256- and 384-bit	never (one failed attempt)	host-verified; protocol pre-registered
ECC2K-95 kernel	$\mathbb{F}_{2^{97}}$, polynomial basis	never	emulated, static
Semaev S_4 sweep; FES	$\mathbb{F}_{2^n}$; $\mathbb{F}_2$	never	host-timed; emulated
sparse matrix–vector	$\mathbb{Z}/m$, $m < 2^{63}$	never	CPU-rechecked
64-bit prime-field rho (sibling)	$\mathbb{F}_p$, $p < 2^{64}$	never	emulated, static

different kinds of evidence and the rest of this paper has kept them apart.

7.1 What has and has not run on a GPU

Table 8 is the inventory. Every "never" is a sentence in the family's own README, and in each case the reason is the same: the code was written and verified in containers without a GPU or a complete CUDA toolkit, and the device runs were planned for rented hardware that the campaign's account hold (Section 9.5) or missing credentials then kept out of reach.[102][103][104][105][106] The numbers below are therefore of three kinds, and each is labelled: a static count, an emulated operation count or walk constant, or (in Section 7.7) a device measurement.

[102] `gpu/ecc/README.md`.
[103] `gpu/btcpuzzle/README.md`.
[104] `gpu/ecc2k/README.md`.
[105] `research/nist_gpu_arithmetic_20260926/ATTEMPT.md`.
[106] `cryptanalysis:docs/GPU.md, "How this is tested"`.

Table 9: Static PTX instruction counts for one 256-bit field multiplication (clang, `sm_90`, `-O3`). The hard floor for the product is 128 instructions: 64 limb products at two instructions each. Class: engineering; nothing here has executed on a device.

	portable C++	inline PTX, before	inline PTX, after
512-bit product	236	171	137
secp256k1 reduction	146	130	77
full multiply	472	328	**210**
Montgomery CIOS multiply	475	—	352

7.2 256-bit prime-field arithmetic for secp256k1

The field layer holds an element in eight 32-bit limbs, because the integer datapath is 32 bits wide and the carry-chain instructions `mad.lo.cc`, `madc.hi.cc` and `addc` exist only at that granularity. For secp256k1 the 512-bit product is folded with $2^{256} \equiv 2^{32} + 977 \pmod{p}$, about 40% fewer multiply-adds than Montgomery; for any other odd modulus below 2^{256} a Montgomery CIOS path is used, and the scalar ring modulo n reuses it on the host. Points are Jacobian with the standard $a = 0$ doubling ($2M + 5S$) and mixed addition ($7M + 4S$) formulas. The library is a throughput library, not a signing library, and is not constant-time.[107][108]

Table 9 is the instruction ledger. The portable path emits no carry instructions at all: of the 444 instructions in a portable squaring, 178 are adds, 102 shifts, 100 ANDs and only 46 multiplies, so the multiply is carry-bound, not multiply-bound, and the inline-PTX rewrite's 55% cut comes from the carry chains. The seven assembly blocks are checked by an interpreter against the portable path on 20,000 random and boundary inputs each, and that check is mutation-tested; it does not cover register allocation or device semantics, which is why `FP_PTX` is off by default.[109][110]

In the unit of Section 4, a secp256k1 inversion is 270 multiplications (255 squarings and 15 multiplications along the libsecp256k1 addition chain), a rho step with batch W is $6 + 270/W$ multiplications (3 for the affine addition, 3 for Montgomery's trick), $7 + 270/W$ with the $j = 0$ fold: 39.8 at $W = 8$, 14.4 at $W = 32$, 8.1 at $W = 128$, against 276 unbatched.[111] Three things were tried and declined on the static count: a dedicated squaring routine (−2.7% portable, +23.5% on the PTX path; reverted), lazy reduction to $[0, 2^{256})$ (worth about 10% of a multiply, but it breaks the canonical form that equality, the distinguished-point test, the cycle hash and the fold comparison rely on), and Karatsuba (judged to lose on a carry-bound multiply). Tensor cores were considered and not reached for.[112] Marking the inversion and ladder routines `__noinline__` and replacing two 16-entry tables by a small double-scalar ladder took the walk kernel's stack from 4,664 to 1,616 B (the low-memory variant from 4,008 to 792 B) and the compile from over ten minutes to 46 s, the same code-size lesson as Section 4.1.[113]

7.3 Rho with automorphism folding, and baby-step giant-step

The rho walk is r-adding on the low bits of x, with distinguished points by a mask on the next byte and three folds: identity, the negation map, and on $j = 0$ curves the full $\langle \beta, -1 \rangle$ orbit of order six, chosen by minimising x over $\{x, \beta x, \beta^2 x\}$ at the cost of one multiplication ($\beta^2 x = -(x + \beta x)$) and two compares, about 10% of a batched step for $\sqrt{3}$ fewer steps. Only an

[107] `gpu/ecc/fp256.cuh`.
[108] `gpu/ecc/point.cuh`.
[109] `gpu/ecc/OPTIMIZATION_BLACKWELL.md, Â§2`.
[110] `gpu/ecc/ptx_asm_check.py`.
[111] `gpu/ecc/OPTIMIZATION_BLACKWELL.md, Â§1`.
[112] `gpu/ecc/OPTIMIZATION_BLACKWELL.md, Â§2 and Â§4.4`.
[113] `gpu/ecc/OPTIMIZATION_BLACKWELL.md, Â§3`.

Table 10: Rho and baby-step giant-step on a 40-bit toy curve ($n = 649{,}523{,}094{,}257$), host emulation of the device source, 128 chains, mean over 8 random targets. S = group additions/$\sqrt{n}$ with table construction charged. The rho rows predate the order-six fold and were single solves ($\pm 0.5\times$). Twelve targets on a 2^{26} interval cost $S = 0.560$ per target against 2.131 for one cold solve.

row	table/$\sqrt{n}$	mean S	predicted	correct
rho, negation map (reference)	0	0.85	—	1/1
rho, no negation map	0	1.42	—	1/1
BSGS textbook, reference stepper	0.707	1.44	1.41	21/21
BSGS textbook, batched $W = 16$	0.707	1.44 (bit-identical)	1.41	21/21
BSGS negation layout, $W = 16$	0.500	1.07	1.00	21/21
giant phase only	0.500	0.56	0.50	8/8

automorphism code is kept per step; a report is $\{x, \text{walk}, \text{restart}, \text{steps}\}$, 48 bytes, and the host replays the two colliding walks from their seeds, multiplying (a, b) by $(-1)^{\text{neg}}\lambda^k$ at each step. Fruitless cycles of length 2 to 6 are detected with a five-deep ring and escaped by doubling the member with the smallest hash; their measured rates at $R = 8$ agree with the prediction to about 10%.[114]

On a 36-bit $j = 0$ toy curve in host emulation (8 solves per fold, 16 walks, $R = 256$, period 2^7), the mean steps were 305,740, 199,042 and 113,522 for folds 1, 2 and 6 against predictions of 249,454, 176,391 and 101,839 ($1.23\times$, $1.13\times$, $1.11\times$; $\pm 0.18\times$ on a mean of eight), all 8/8 correct, so fold 6 buys $1.75\times$ over the negation map (predicted 1.73). In the unit S = ops/$\sqrt{n}$ that is 1.54, 1.00 and 0.57. The three steppers (batched, low-memory, one-inversion-per-walk) are bit-identical over hundreds of iterations with the same distinguished-point multiset, and every point replays on the host to $aP + bQ$, which is what proves the λ bookkeeping on secp256k1 itself.[115]

Baby-step giant-step on the same arithmetic keeps an x-keyed table of 16-byte entries in lock-free open addressing (one 64-bit `atomicCAS` per probe, about 2.5 slots per probe at half load, a false-candidate rate of load/2^{32} and none observed), and walks W giant chains per thread sharing one inversion, 94 registers and no spills at $W = 8$. With the negation map the layout is $m = \sqrt{w}/2$ babies and $2m - 1$ giants, the Galbraith–Wang–Zhang row at $1.00\sqrt{w}$ expected against the textbook $1.41\sqrt{w}$. A 2^{66}-wide interval is 2^{32} entries and 64 GB, the top of one 80 GB device; beyond that the method is out of memory, not out of time.[116] Table 10 is the measured cost on a 40-bit toy curve, in emulation, every phase charged.

The classification is the same as every row in Section 5: the batched stepper is engineering, the layouts are the published ones, and nothing moves the ratio to the generic floor. A Rust port of the same baby-step giant-step on the CPU runs at 37–49 M steps/s on four threads at $S = 1.03$ cold and 0.55 amortised, 6–11× the big-integer implementation it replaced.[117] The emulation itself runs at about 380 k steps/s per core, a figure that says nothing about a device and is quoted only so that no one mistakes it for one.

7.4 Pollard kangaroo for the interval problem

The interval solver reuses the verified field and point code for $Q = kG$ with $k \in [a, a+w)$ at about $2\sqrt{w}$ steps: two herds that alternate by thread parity so every warp is half tame and half wild, a jump table of 2^6 pseudo-random scalars with mean jump $\sqrt{w}/2$ and 25-word entries (odd, so

[114] `gpu/ecc/README.md, "The walk".`
[115] `gpu/ecc/README.md, "How this is tested".`
[116] `gpu/ecc/README.md, "Baby-step giant-step".`
[117] `gpu/ecc/README.md, "Cost, measured".`

the shared-memory reads are bank-conflict-free), and the target centred at $Q' = Q-(a+w/2)G$. The invariant that makes it testable is that after every step a tame kangaroo at distance d is at dG and a wild one at $Q' + dG$, which pins the jump table, the distance accumulator and the point arithmetic together.[118]

Its measured constant is the useful number. In emulation on a 2^{31} interval with 120 trials per configuration, steps are $C\sqrt{w}$ with $C = 2.79 \pm 0.13$ for the default (median 2.65); the idealised two-herd bound is 2.0, Pollard's classic lambda about 3.3, and published parallel implementations about 2.1, so this one is about a third above the best published. A jump-parameter sweep at 20 trials per cell had shown C from 2.50 to 3.95, an apparent 36% win; rerun at 120 trials the best cell and the default were within half a standard error. The defaults sit on a flat optimum and the sweep was noise, which is recorded so that it is not rediscovered. Restarting at a distinguished point or not was likewise within noise.[119] A multi-device split (one host thread per device, disjoint even index bases so herd parity stays global, one host table) is tested by reproducing the combined herd exactly from two contexts and recovering a 28-bit key across them; multi-machine is not done.[120] The tool ships no real puzzle keys (no trustworthy offline source), does no address scanning, and has not solved any puzzle; the synthetic 20-, 28- and 32-bit registry entries are regression fixtures.

7.5 P-256 and P-384

A separate library targets the NIST primes on the RTX PRO 6000 with 8 and 12 limbs, thread-private elements, CIOS Montgomery with $n' = 1$ (both primes have low limb $2^{32} - 1$) and a multiply-free reduction row (every modulus limb is 0, 1, $B - 1$ or $B - 2$), against a matched generalised-Mersenne competitor with two fold rounds. It is verified on 20,000 random pairs per curve and representation against Boost.Multiprecision, with Jacobian doubling and mixed addition checked against independent affine arithmetic, and its benchmark protocol was pre-registered: workloads, batch size, seed, sweep, three repeats and a 5% threshold below which the simpler configuration wins by rule.[121][122] The one device attempt is an infrastructure non-result: the cloud credentials were not exposed to the CI runner and the GPU steps were skipped. The attempt record says it "must not be cited as performance evidence", and it is not.[123]

7.6 Binary fields beyond ECC2K-130

The ECC2K-95 kernel. Section 2.3 used this kernel as the end-to-end control; its arithmetic is the packed, polynomial-basis counterpart of Section 4.2. An element is four 32-bit words; a 32-bit carry-less product is sixteen widening integer multiplies with the interleaved-mask trick (operands split by bit index modulo 4 so no partial sum exceeds 8 and nothing carries); the 1-bit top word at $m = 97$ is split off so that a three-way Karatsuba over three words (six leaves) plus seven single-bit masks replaces nine leaves; reduction by $t^{97} + t^6 + 1$ is two folds, the second an extract and two XORs; squaring is a bit-spread kept in 64-bit form (78 instructions against 104 for two 32-bit spreads); inversion is Itoh–Tsujii at about 7 multiplications and 96 squarings. The static count is 304 instructions per multiplication, down from 405, with the six software leaves 96 of the 304; the native `clmad` that gave the ECC2K-130 client +22% would do the same product in six instructions and is not yet ported because the pip-wheel toolchain stops at `ptxas` 12.9.[124][125]

[118] `gpu/btcpuzzle/README.md`, "How the walk works".
[119] `gpu/btcpuzzle/README.md`, "Measured cost".
[120] `gpu/btcpuzzle/README.md`, "Multi-GPU".
[121] `gpu/nist/README.md`.
[122] `research/nist_gpu_arithmetic_20260926/PROTOCOL.md`.
[123] `research/nist_gpu_arithmetic_20260926/ATTEMPT.md`.
[124] `gpu/ecc2k/README.md`, "What the field arithmetic actually costs".
[125] `gpu/ecc2k/f2m.cuh`.

Per step the cost model is one inversion split W ways plus two multiplications, one squaring and the three multiplications of Montgomery's trick, with the two Frobenius chains of the walk paid in squarings: 3,813 instructions at $W = 8$, 44% multiplies, 27% the Frobenius chains, 25% the inversion's squarings. Because τ^k is $\mathbb{F}_2$-linear it can be applied as 25 four-bit-window table reads at any k, 296 instructions against 78 per squaring; tables for $\tau^{6,12,24,48}$ cut the inversion's 3,744 squaring instructions at $k = 48$ to 296 and the step to 3,083 (-19%), and $W = 4$ with tables beats $W = 8$ without at 656 against 1,048 bytes of local memory. The table traffic, 400 words per inversion, is unmeasured, and the tables are off by default.[126] The low-memory stepper at $W = 8$ compiles to 128 registers and 464 B of stack with four blocks per SM, 512 threads per SM on `sm_90`, `sm_100` and `sm_120`; the prime-field walk of Section 7.2 needs 226 registers and 792 B at 256 threads per SM, so the binary walk gets 1.5–2× the occupancy before either has run.[127] A 40-bit end-to-end test of eight logarithms on one CPU core fell from 2.1–2.7 s to 1.02–1.05 s across the rewrite at the same 191,840 steps, which the README itself files as engineering with S flat.

The pair table. For Koblitz index calculus the same directory builds the meet-in-the-middle table of all $P_i + P_j$ over a factor base, one thread per row with the row's denominators batch-inverted, folded to signed Frobenius orbits by the least cyclic rotation of x in a normal basis (one representative times an orbit-sorted suffix, $2n$ times fewer sums). Stored words shrink 43.4× at $n = 23$ (6,256 against 271,216) and 77.2× at $n = 41$ (11,152 against 861,328). Verification runs each emulated thread on its own CPU thread, requires identical tables at several thread counts and chunkings, round-trips every stored pair and 19,999 generator multiples into the Rust table, keeps a control that must lose pairs, and runs under ThreadSanitizer with a deliberately racy control that must be reported; the bug that motivated all this (the base size passed as the field degree) fails 960 of 1,176 keys when restored.[128][129] The fold does not reach $n = 97$ (limited to $n \le 62$) and the device round trip is planned, not run.

Polynomial systems. Two more kernels serve the decomposition step of Section 10.1. A Gray-code fast exhaustive search over quadratic Boolean systems (one thread per sub-cube, $n \le 40$, ≤ 64 equations, shards checked against brute force) exists alongside the production Rust path and has not run on a device.[130] A Semaev S_4 pairs-and-solve sweep (fix X_1, X_2, reduce the quartic in X_3 against the linearised polynomial of the factor-base subspace, $O(l)$ field operations per pair instead of 2^l) has a `clmad` path and a software fallback, is tested against exhaustive triple enumeration, and on one host core went from 5.07 to 0.44 ms per 2^l-row sweep at $n = 21$. Its README's conclusion is the one that matters: relation collection stays $\Theta(2^n)$ with any oracle polynomial in the base, "the oracle got roughly a thousand times faster and the attack's exponent did not move"; a GPU is another constant. The same README records and retracts an earlier claim that NVIDIA has no carry-less multiply.[131]

Linear algebra. A batched dense reduced-echelon solver over $\mathbb{F}_p$ ($p = 2083$, 226×286 matrices of rank 222) serves the $E(\mathbb{F}_{p^3})$ index calculus whose Macaulay step is 57% of each solve. One archived pipeline experiment did run on an RTX PRO 4500: 64 matrices per invocation, 36 runs, 2,340 matrices verified against the serial reduction, device reduction about 0.045 s per batch inside a 0.32 s cold invocation dominated by setup, and a two-slot overlap 5.1% faster on held-out seeds, which the record declines to call a speedup for want of an exclusive host and an

[126] `gpu/ecc2k/README.md, "$\tau ^k$ is not $k$ squarings"`.
[127] `gpu/ecc2k/README.md, "Occupancy"`.
[128] `gpu/ecc2k/README.md, "The $|F|^2$ pair table"`.
[129] `gpu/ecc2k/pairtable.cuh`.
[130] `gpu/fes/README.md`.
[131] `gpu/semaev/README.md`.

interval. Its own conclusion is that a GPU changes the crossover with rho "by not a single bit"; it would only enable larger p.[132][133] (The directory README's statement that no kernel has run on a GPU predates that experiment.) A sparse matrix–vector worker modulo any $m < 2^{63}$ is correctness-first, every product rechecked on the CPU, with no rate.[134]

7.7 What did run: F4 elimination on rented GPUs

The sibling library's F4 Gröbner-basis path for Weil-descent systems is the one piece of index-calculus arithmetic with device measurements. On a rented RTX 5090 (`sm_120`, 32 GB) the degree-6 Macaulay matrices are built and eliminated 25 to 75 times faster than on the Xeon host; the $168{,}363 \times 188{,}203$ matrix went from 426 s to 5.7 s, the degree-of-regularity sweep from 18.8 to 8.2 s, and the batched per-node decider was correct on the device and 4.4× slower than the host's default engine.[135] On an H100 a tiled update to the elimination kernel was kept at 1.11–1.49× and a merged-basis panel reverted at 1.5–5.3× slower; the largest matrix fell from 4,781 to 3,586 ms.[136] The records call these what they are, stage diagnostics of one oracle matrix, not an ECDLP cost, and a later crossover harness keeps its qualified speedup null until a matched device-and-host resource receipt exists.[137]

The sibling's own 64-bit prime-field CUDA rho (one limb, Montgomery with $R = 2^{64}$, eight walks per thread sharing a Fermat inversion so that a step is 18 multiplications instead of 99) is in the same state as the kernels above: 77,000 device-arithmetic assertions against the host library, whole-solver emulation on four groups, a cubin for `sm_70` through `sm_90` with 64–72 registers and no spills, and no execution on a device; its emulator rows sit at $S \approx 1.3$–2.6 alongside the CPU rho and show only that S does not grow with the walk count.[138][139] Its efficiency scorer, built for the ECC2K-130 walk, is the one place the two threads meet: charging the packed walk's 4,328 lane-instructions per iteration to the measured dual-issue rate of the RTX PRO 6000 (78.3 per SM-clock, 188 SMs, 2,307 MHz) gives an attainable 7.85 B/s per card against an audited 6.905 for that build, 88% of the issue roof, and the same method on the wide multiply reproduces the "31 B/s if the walk did nothing else" of the earlier ceiling note.[140][141]

7.8 What the inventory says

Two lessons carry across the fields. First, the carry-less unit is the whole difference between the binary and prime-field kernels on this hardware: a 131-bit binary product is six `CLMAD` and a 256-bit prime-field product is 210 carry-chained integer instructions at best, so the binary walk is pipe-bound on a narrow unit at 1.6–2.0 lanes per SM-clock while the prime-field walk is carry-bound on the wide integer pipe, and the two want different occupancies, different batch sizes and different arithmetic representations. Second, every one of these libraries was taken to a verified, bit-identical, statically-accounted state without a device, and that state is worth exactly what Section 5.1 says a pre-tuning floor is worth: it bounds what a device could do, and it does not report what one did. The ECC2K-130 client is the only one of them that has been carried through to the measured side of that line, which is why it is the subject of this paper and the others are a section.

[132] `gpu/macaulay/README.md`.
[133] `gpu/macaulay/benchmarks/pipeline-20260914/RESULTS.md`.
[134] `gpu/spmv/README.md`.
[135] `cryptanalysis:experiments/f4-gpu-runpod-20261006/RESULT.md`.
[136] `cryptanalysis:experiments/f4-gpu-panel-20261007/README.md`.
[137] `cryptanalysis:experiments/f4-gpu-query-20261008/README.md`.
[138] `cryptanalysis:docs/GPU.md`.
[139] `cryptanalysis:docs/BENCHMARKS.md, "GPU rho kernel"`.
[140] `cryptanalysis:gpueff/README.md`.
[141] `cryptanalysis:gpueff/profiles/ecc2k130-packed-walk.json`.

8 The FPGA engine

The field is a better fit for an FPGA than for a GPU: the GPU has one narrow carry-less unit per SM, while the FPGA is nothing but carry-less multipliers, and σ^k is wiring. A VHDL engine for the AWS F2 (Xilinx VU47P) was designed, estimated before any board time, and then measured.

8.1 Estimate before the board

Before synthesis, the client's own straight-line multiplier was mapped to LUT6s (about 5,563 at 32 levels, known to be about 40% high for ignoring dual-output packing) and calibrated against the 3,071 and 3,620 LUTs that Bernstein et al. report for GF(2^{113}) and GF(2^{127}) multipliers [7], extrapolated to 3,781 at $m = 131$. With multipliers at 79% of a core and 5.33 multipliers per core, a core is 25.5 KLUT at literature packing or 37.5 KLUT with the repository's unpacked circuit; 18% shell overhead on 1,303,680 CLB LUTs at 70–80% utilisation gives 29–34 or 20–23 cores at an estimated 250–350 MHz. The estimate was 5–12 B iterations/s per VU47P, central 8 B/s: slower per chip than the 14.1 B/s GPU, about level on cost per solve, and 2–5× better per watt.[142]

8.2 Design

Multiplier. `gf131_mul` follows the Bernstein–Lange route in hardware: `prep` converts each operand to the polynomial basis (991 XOR2 per operand), a recursive Karatsuba `gf2_kmul` splits $131 \to 66 \to 33 \to 17$ bits into 27 leaf products of 17×17, and `to_onb` converts the 261-coefficient product back (3,167 XOR2). There is no reduction step. At three levels the product is 7,803 AND and 9,404 XOR2, 14,553 XOR2 per multiply including conversions, against 17,161 AND and 22,049 XOR2 for schoolbook. Out of context in Vivado 2025.2 at a 4.0 ns clock, three levels map to 4,855 LUTs and 4,647 FFs with +2.68 ns of slack.[143]

A DSP leaf computes a 17-bit GF(2) product as six 25×16 integer products with coefficients spaced three bits apart (split $9 + 8$ against $6 + 6 + 5$) so that no column count exceeds six and nothing carries; the parities are XORed into place with 22 LUTs. Eleven of the 27 leaves go to DSPs, 66 DSP48E2 per multiplier, which takes the multiplier to 4,004 LUTs at a latency of 14 clocks including a 261-FF copy register (11 clocks with LUT leaves). The pblock holds 7,992 DSPs, so 120 engines can take 11 DSP leaves and 128 engines ten.[144][145]

Step and batch. An unbatched step is ten multiplies: $d = x + \sigma^j x$, $\lambda = (y + \sigma^j y)/d$ with the eight-multiply Itoh–Tsujii inversion, $x_3 = \lambda^2 + \lambda + d$, $y_3 = \lambda(x + x_3) + x_3 + y$. `ec2k_batch_pipe` shares one inversion across W walks through a product tree held in UltraRAM (four URAM288 per engine): $W - 1$ forward multiplies, 8 for the inversion, $2(W - 1)$ backward, W for λ and W for y_3, $5W + 5$ per batch, $5 + 5/W$ per step: 5.31 at $W = 16$, 5.16 at $W = 32$, 5.08 at $W = 64$. A batch is a chain of $2 \log_2 W + 10$ bursts, and several batches are kept in flight to hide the multiplier latency. Table 11 is the simulated rate.[146][147]

Walker and host port. Walks live in a block-RAM FIFO of $(\mathrm{id}, \mathrm{steps}, x, y, \mathrm{dp})$, 285 bits by 512, read through a four-entry prefetch. Only the low 13 bits of the step count travel with the walk; the high 19 sit in a LUTRAM table bumped once every 8,192 steps. A step whose x

[142] `ecc2k130/FPGA-CEILING.md`.
[143] `hdl/ecc2k130/README.md`.
[144] `hdl/ecc2k130/gf2_dsp_leaf.vhd`.
[145] `hdl/ecc2k130/README.md`.
[146] `hdl/ecc2k130/ec2k_batch_pipe.vhd`.
[147] `hdl/ecc2k130/README.md`.

Table 11: Simulated clocks per step of the batched step unit at multiplier latency 10, against the arithmetic bound $5 + 5/W$. The default geometry is 64×8 (512 walks per engine), which runs at 5.16 clocks per step with either multiplier.

W	batches in flight	200 steps	6,144 steps	bound
8	8	8.02	6.62	5.63
16	4	9.86	8.38	5.31
16	8	6.98	5.67	5.31
16	16	6.46	5.34	5.31
32	8	7.20	5.20	5.16

Table 12: F2 images on one VU47P (`f2.6xlarge`). All runs had zero dropped reports; for every image except the 112- and 120-engine ones, 32 or 64 sampled distinguished points were re-walked on the host against the client reference from the same seed and matched. The 128-engine image is the promoted AFI.

engines	MHz	geometry	routed WNS (ns)	M steps/s	clk/step
48	333.3	16×16, tree in BRAM	+0.022	3,010.7	5.31
64	333.3	same	+0.040	4,013.7	5.32
80	333.3	17 tiles/engine	+0.027	5,015.8	5.32
96	333.3	same	+0.002	6,018.0	5.32
112	375	32×8, tree in URAM	+0.049	8,123	5.17
120	333.3	32×8, tree in URAM	+0.063	7,730	5.17
128	**333.3**	32×8, tree in URAM, LUT multiplier	+0.048	**8,249**	**5.17**

has weight at most `DP_WEIGHT` is flagged, moves into a report register at the FIFO head and is held until acknowledged, so no report is ever lost and the `DROPPED` counter always reads zero. `ec2k_axil` is an AXI4-Lite slave in front of `NENG` walkers with a credit-based register spine (about 870 FFs and 300 LUTs per engine), introduced after the first flat 48-engine build placed at -2.5 ns. The host program is a drop-in for the CUDA client under the same worker script: it writes the same 32-byte records and can re-walk the first N reports from their seeds on the CPU (`-verify N`); production workers leave verification to the campaign merge.[148][149]

Verification. The reference script imports the client's field model and emits 256 multiplier vectors (including 0, the all-ones 1, single basis elements and the extreme coordinates, each also checked against an independent direct convolution), 200 steps (twelve full batches of 16 plus a partial batch, so the flush path runs every time) and 64 walks of random order-ℓ points on the challenge curve. The batch testbench passes for every W from 2 to 32 with 2 to 16 batches in flight; the walker testbench checks every report's id, step count and point against the reference trajectory (64 distinguished points from 797 steps in 19,194 clocks).[150][151]

8.3 Measured on `f2.6xlarge`

Table 12 is every image that was built, placed, loaded and timed, and Figure 1 draws it.[152] The 48-engine run read back its geometry (48 engines of 512 walks, weight 34, 333.3 MHz), seeded 24,576 walks over AXI-Lite in 7.3 s, held 3,010 M steps/s from ten seconds on, and completed 103 G iterations in a 40 s run; simulation had predicted 5.29 clocks per step against

[148] `hdl/ecc2k130/ec2k_walker.vhd`.
[149] `hdl/ecc2k130/host/ecc2k130_fpga.cpp`.
[150] `hdl/ecc2k130/ecc2k_ref.py`.
[151] `hdl/ecc2k130/vectors_ecc2k130.txt`.
[152] `hdl/ecc2k130/aws/README.md`.

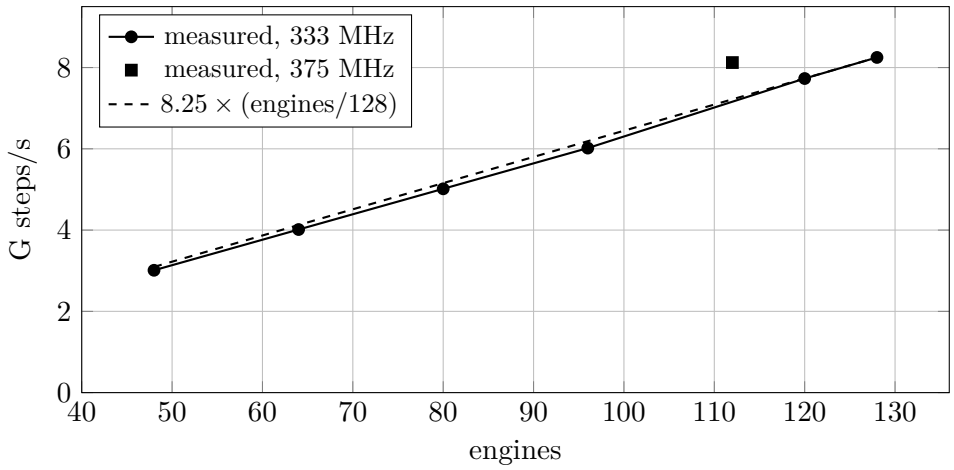


Figure 1: ECC2K-130 FPGA engine on AWS F2. The 128-engine image at 333 MHz is 8.25 G steps/s; the pre-board estimate was 5–12 G with a central 8 G. The measurement landed on the central estimate and is a device number, not the estimate reprised.

the measured 5.31. The promoted 128-engine image holds 8.25 G steps/s. Linear scaling with engine count is the prediction of a design that shares nothing between engines except the host port, and the measured images sit on it. Builds that missed timing (128 engines at 375 MHz, 112 at 400) and the 144-engine build that failed block-RAM placement (3,744 RAMB18 sites needed against 3,576) are recorded alongside; the projection at 5.16 clocks per step is 8.8 G at 136 engines and 9.3 G at 144 or at 128 engines and 375 MHz, neither yet met (128 at 375 MHz routed at −0.21 ns; 144 failed placement).[153]

Per engine the current revision is 6,790–6,940 LUTs, 9,371 FFs, 12 RAMB36, 4 URAM and 66 DSPs; the 96-engine image used 62% of the LUTs and 81% of the block RAM, and block RAM, not logic, is what bounds the engine count on this part.[154]

8.4 F2 against the GPU

At the prices observed on 2026-09-11, an F2 instance at $1.980 on demand or $0.709 spot against an RTX PRO 6000 at $4.14 and $1.31 per GPU-hour, the expected $2^{60.9}$ iterations cost about 72,400 F2-hours at the measured 8.25 G/s, roughly $51k spot, against 42,400 GPU-hours and $56k spot at the campaign's 14.11 B/s collection rate.[155] The two platforms are therefore level on cost per solve within the noise of spot pricing, and the FPGA is about 2–5× better per watt on the note's estimated envelopes (75–150 W against a 600 W board). FPGA workers speak the same client contract as the GPUs and write the same records, so a collision between an FPGA trail and a GPU trail is an ordinary merge.

9 The campaign

$2^{60.9}$ iterations is about five years of single-GPU time at the campaign's collection rate, so the campaign is collection, not a solve: a corpus of distinguished points, each re-walkable from its seed, grown by whatever hardware is cheapest at the time and merged on arrival. This section records its configuration, its calibration, its storage protocol and its state.

[153] `hdl/ecc2k130/README.md`.
[154] `hdl/ecc2k130/README.md`.
[155] Derived here from the measured rate and the prices recorded in `ecc2k130/FPGA-CEILING.md`; the note's own table priced the hypothetical 8 B/s image at $53k spot.

9.1 Configuration

The configuration is one file in the repository; the copy in the storage bucket is what workers read.[156][157] Walk: σ (Section 3.1). `dpWeight`: 32, uniform campaign-wide, and a campaign run refuses any other. Batch 16, 256-thread blocks, two resident blocks per SM, 385,024 workers = 6,160,384 logical walks per GPU, 1,024 steps per launch, checkpoint and upload every 600 s, restart every 48 h. The iteration guard `maxIters` in the repository file is 2^{32}, twelve trail lengths at weight 32; it was 2^{30}, three trail lengths, which a model on the measured rates says cut 4.9% of walks and discarded 15.6% of the fleet's steps until an audit found it. Whether the bucket copy was raised before the fleet stopped is not recorded.[158] The fields that define the corpus (workers, batch, geometry, curve, cutoff, walk and the state-layout knobs) freeze once any slot exists; `WITNESS=0` is pinned. The preset is the audited RTX PRO 6000 build at CUDA 13.3.1 and `sm_120`.[159] A checkpoint is about 370 MB (six million walks at 60 bytes).

9.2 Cutoff 32 and the interval $2^{28.41}$

The planning numbers follow from the cutoff.[160] At weight 34 the expected corpus is $2^{35.6}$ records, about 1.7 TB, one per $2^{25.27}$ iterations; at weight 32 one report per $2^{28.41}$ iterations, $2^{3.14} = 8.8\times$ rarer, so about $2^{32.5}$ records and 0.19 TB at the expected work, about 40 points/s, 3.4 M points/day and 0.11 GB/day from a GPU collecting at 14.11 B/s. The price of a rarer point is the work left in flight when a collision lands: 43 GPU-hours per GPU at weight 32 (6.16 M walks times $2^{28.41}$ steps), 13% of the expected work at 128 GPUs and more than the whole expected work at 1,024, against 4.9 GPU-hours per GPU at weight 34. The campaign's fleet size never approached the range where that trade reverses.

The interval itself was measured twice, and the first measurement was wrong. A fleet-wide ratio of $7.93 \cdot 10^{15}$ checkpointed iterations to 31.6 M uploaded points on 2026-09-15 had been quoted as $2^{27.9}$ iterations per point. Neither counter was clean: the checkpoint sum multiplied every slot's per-walk base by the Blackwell grid constant, inflating the auto-sized Ada slots by the ratio of the grids (about 26× on an L4-sized grid), and the point total mixed in two retired slots that had walked at weight 34. The replacement reads from each live worker the client's own two counters, iterations this run on the grid the client built itself and points reported over the same interval, frozen at 2026-09-17 18:47 UTC across 132 workers (Table 13).[161] The result, $2^{28.41} = 3.56 \cdot 10^8$, is the same on every hardware family to 0.04 in the exponent, and sits $2^{0.60}$ below the naive binomial $P(\mathrm{HW} \le 32) = 2^{-29.01}$, as Bailey et al.'s $2^{25.27}$ sits $2^{0.57}$ below the binomial at weight 34. The correction is classified as accounting: the walk did not change, the number did, and the earlier figure is retired rather than averaged.

9.3 Fleet economics

Cost is a property of the work, not of the fleet size: at the 2026-09-11 spot price of $1.31 per RTX PRO 6000 hour the expected 42,400 GPU-hours are about $56k whether 8 GPUs take 221 days or 512 take 3.4; on demand the figure is $142k–$175k.[162] What the fleet size buys is the collision-in-flight loss above and the calendar. The campaign ran mixed families because iterations per dollar, not iterations per card, is the objective: on 2026-09-18 the best-value quote was an L4 in `eu-north-1` at $56 \cdot 10^{12}$ iterations per dollar against $38 \cdot 10^{12}$ for the RTX PRO 6000

[156] `ecc2k130/aws/campaign.json`.
[157] `ecc2k130/aws/README.md`.
[158] `research/ecc2k130_rho_audit_20260925/README.md`.
[159] `ecc2k130/aws/README.md`.
[160] `ecc2k130/aws/README.md, "The numbers that decide the plan"`.
[161] `ecc2k130/benchmarks/dp-interval/README.md`.
[162] `ecc2k130/aws/README.md`.

Table 13: Iterations per distinguished point at cutoff 32, frozen 2026-09-17 18:47 UTC. Per-worker spread 0.04 in the exponent. The last column is the median self-reported rate of the family's workers on the same day.

family	workers	iterations	points	it/point	median B/s
RTX PRO 6000 (`g7e`)	38	$5.606 \cdot 10^{15}$	15,724,672	$2^{28.41}$	14.66
L40S (`g6e`)	34	$2.723 \cdot 10^{15}$	7,643,600	$2^{28.41}$	7.47
RTX PRO 4500 (`g7`)	48	$3.081 \cdot 10^{15}$	8,640,973	$2^{28.41}$	4.87
CPU	12	$2.804 \cdot 10^{14}$	785,410	$2^{28.41}$	2.29
all	132	$1.169 \cdot 10^{16}$	32,794,655	$\mathbf{2^{28.41}}$	$\approx$ 1,068 total

and $28 \cdot 10^{12}$ for the RTX PRO 4500, and at that quote a full reference walk costs about $38k.[163] The per-SKU rates behind those ratios are measured, not list: L40S 8.84 B/s with `clmad`, L4 2.49, T4 0.53 without it.[164][165]

The RTX PRO 4500 is the instructive case. Its measured 5.107 B/s on the shipping walk is exactly what the 6000's rate predicts from SM count and clock ($14.64 \times 82/188 \times 1.95/2.44 = 5.10$), and at that rate the 6000 wins per dollar by 1.2–1.3×; the break-even against the 6000's shipping rate is 6.2–6.6 B/s in the two cheapest regions and 8.8 in the third, figures the note says must be redone with both cards on their best builds.[166] That is why a 12 B/s goal was set for the card and why the x-only bridge map of Section 3.3 was pursued there rather than on the 6000: a power-capped die rewards fewer instructions per step more than a pipe-bound one does. The separate goal of 28 B/s on one 6000 was priced below the pipe (Section 5.1) and its grid and occupancy screens, run on an RTX PRO 4500 because no 6000 could be allocated, found nothing faster than the shipping configuration.[167]

A power survey closes the loop: on the 6000 the 20 B/s build drew 583 W for 34.4 M updates per joule and a fused-pass variant 544 W for 37.0 M/J in the same session, 1.072× per joule in five paired repetitions against a 1.03 target; per joule, not per second, is the figure that transfers to a 165 W card, and the model, calibrated on the 4500's 5.107 and 5.552 B/s receipts (ratio error +2.6%), predicts 7.2–7.4 B/s for the fused build there.[168] That prediction is owed a measurement; the bridge map arrived first.

9.4 Storage, seeds and certification

A record is 32 bytes: a little-endian 64-bit seed and the canonical normal-basis x of the endpoint, three 64-bit words; no scalar coefficients are stored, they are rebuilt by replay. The seed is `run id (16)` ‖ `walk index (32)` ‖ `restart counter (16)` with run id = slot +1; a splitmix64 finaliser expands it to the 128-bit c of the start point. Two contributors who pick different run ids never start the same walk (independently seeded walks can still meet, which is the point), and a permanent seed registry does compare-and-set across providers (Modal GPU runs are ids 8000–8999, CPU sidecars 9000–9999); a race test with eight claimants acquired exactly one.[169][170][171][172]

Uploads are ordered so that no pointer ever names data that is not there: payloads before

[163] `ecc2k130/aws/RESEARCH-CREDITS.md`.
[164] `ecc2k130/ADA-L4-L40S.md`.
[165] `ecc2k130/T4-G4DN.md`.
[166] `ecc2k130/RTX-PRO4500.md`.
[167] `ecc2k130/GOAL28-RESULTS.md`.
[168] `ecc2k130/POWER-BOUND.md`.
[169] `ecc2k130/CERTIFICATION.md`.
[170] `ecc2k130/SEED-IDENTITY.md`.
[171] `research/ecc2k130_rho_audit_20260925/README.md, "Seeds"`.
[172] `ecc2k130/aws/README.md`.

their commit manifests, both before checkpoint publication. Strict manifests bind curve, cutoff, guard, seed and key semantics, worker geometry, source hash, worker binary hash and replay binary hash, and every immutable payload carries its SHA-256. Slot ownership is one S3 object per slot claimed with a conditional put on the ETag, a 60 s heartbeat and a 180 s lease; checkpoints are content-addressed and the owner-checked heartbeat publishes the pointer, so a stale worker can leave orphaned blobs but cannot commit. The note is explicit that this is integrity against failure, not authentication against an actor able to rewrite both data and manifests.[173][174]

Merging buckets records by a hash of the key limbs (the first layout used the key's low bits) into 4,096 files, sorts, drops exact duplicates (the client store, the database primary key and the merge are three layers; 9.95 M re-reports had been dropped by 2026-09-25) and reports equal keys with different seeds. Such a pair goes into a two-record file and to the CPU client, which re-walks both trails, aligns Frobenius and negation, and verifies $[k]P = Q$; a $\mathrm{GF}(2^{41})$ pair with known seeds that recovers $k = 369250562913$ is the local acceptance fixture. An exhaustive check of 76,904,684 step multisets confirms that no multiset of at most 32 σ-walk steps closes an orbit, so no short fruitless cycle exists; a trail crossing itself at random is estimated at about 2^{-65} per trail, about 10^{-10} over the campaign, and the iteration guard releases such a walk.[175][176][177]

The public dashboard publishes aggregates only: not point keys, coefficients, seeds or database locations.[178][179]

9.5 State

The campaign was created on 2026-09-11. Its fleet on 2026-09-17 was about 133 workers (the 132 with enough points to calibrate are in Table 13), producing about 2,600 points/s by the ingest's count; the ingest path was re-engineered that week from one object at a time (2,450 records/s, below production) to a threaded loader measured at 137,753 rows/s, and a five-and-a-half-hour blind spot during a key layout change was recovered in 23 minutes (4,321 objects, 63,351,803 rows, zero failed), taking the store from 67.6 M to 124.3 M points.[180] On 2026-09-18 the spot market reclaimed every worker and the cloud account was placed on an account-verification (risk) hold that has blocked new instance launches since; collection continued at a small scale on another provider. Its first four GPU runs reused the run ids of the first four slots by mistake and re-walked them (813k of one run's 912k records byte-identical to the slot they replayed, all dropped as re-reports, and an inadvertent end-to-end replay check); later runs use the reserved ids 8000–8999.[181][182]

Table 14 gives two snapshots: the audited one of 2026-09-25, while a small fleet was still walking, and the published feed of 2026-10-10, by which time the last point had landed (2026-10-07) and no slot was walking; the sustained peak and the mean trail length in the caption are from the audit.[183][184][185] No collision is the expected state: $2^{56.68}$ iterations is 5.4% of $2^{60.9}$, and the probability of no collision after a fraction c of the expected work is about $\exp(-\pi c^2/4)$,

[173] ecc2k130/CERTIFICATION.md.
[174] ecc2k130/aws/README.md, "Slots".
[175] ecc2k130/CERTIFICATION.md.
[176] research/ecc2k130_rho_audit_20260925/README.md.
[177] ecc2k130/aws/README.md.
[178] docs/ecc2k130-status/README.md.
[179] scripts/rho_status/README.md.
[180] ecc2k130/aws/README.md.
[181] ecc2k130/aws/ACCOUNT-HOLD.md.
[182] ecc2k130/README.md, "Collecting for the campaign from Modal".
[183] research/ecc2k130_rho_audit_20260925/live_status_20260925T050509Z.json.
[184] paper/ecc2k130/data/status_20261010T154157Z.json.
[185] research/ecc2k130_rho_audit_20260925/README.md.

Table 14: Campaign state at the audited snapshot of 2026-09-25 05:05 UTC and at the published feed of 2026-10-10 15:42 UTC. The pooled iterations-per-point ratio is below $2^{28.41}$, partly because the corpus still holds weight-34 records from the first two slots; the mean length of walks that ended by 2026-09-25 is $2^{28.34}$. Sustained production peaked at about 842,000 points per hour on 2026-09-23, about 83 B it/s.

	2026-09-25	2026-10-10
distinguished points stored	367,123,410	485,528,850
iterations (checkpointed)	$2^{56.38}$	$2^{56.68}$
share of expected work	4.4%	5.4%
collisions	0	0
duplicate re-reports dropped	9,953,494	14,153,833
slots / walking / off-weight	195 / 2 / 7	227 / 0 / 7
points in the last 24 h	741,048	0
walk rate over the last hour	3.65 B it/s	0
last point stored	—	2026-10-07 14:41

here 0.998. Collection is paused, not finished: every slot's checkpoint and the corpus are intact, and a worker that resumes a slot continues its walks from the checkpoint.

10 Structural alternatives, measured

Every row above is engineering against a fixed exponent. Three threads asked whether the exponent could move for this curve, in the same accounting, and each is reported here with its measurement because a later reader who rediscovers the idea should find the number rather than the folklore. None changed the campaign.

10.1 Index calculus on the Koblitz curve

Index calculus on $E(\mathbb{F}_{2^n})$ decomposes a random point into m factor-base points through Semaev's summation polynomials [12, 15, 16, 18, 25, 27]; on a Koblitz curve the factor base can be taken Frobenius-stable so that columns are orbits, saving a factor n in relations and n^2 in linear algebra against rho's $\sqrt{2n}$. Two structural facts about $n = 131$ come first. In a normal basis the Frobenius is a coordinate permutation, so a Frobenius-stable subspace is an $\mathbb{F}_2[t]/(t^n - 1)$-submodule, a binary cyclic code of length n, and at $n = 131$ (and $n = 163$) 2 is primitive modulo n, so $t^n - 1$ has exactly two factors and the only stable subspaces have dimension $0, 1, 130, 131$. The usual subspace construction is empty at exactly the interesting sizes; forced onto the 130-dimensional subspace, the orbit trick is a net loss of 57–96 bits for $m \in \{2, \ldots, 5\}$.[186][187] The Couveignes–Lercier elliptic-period alternative needs an auxiliary curve over $\mathbb{F}_2$ with $n \mid \#E'(\mathbb{F}_2)$, and Hasse gives at most 5.[188] What survives is a weight-bounded base $\{R : \mathrm{HW}(x_R) \le w\}$ in the normal basis, which is Frobenius-stable for every n, is the predicate the walk already computes, and is not a low-degree algebraic condition, so only a solver that does not need linearity can use it.

SAT decomposition. Chaining Semaev's S_3 through the intermediate sums, with squaring free, encodes a k-point decomposition as a CNF whose multiplications cost the generated Karatsuba circuit's gates (15,588 at $n = 131$ against 2.2 million for the definition). With $k = 3$, $w = 3$ and CryptoMiniSat the median solve is 0.11 s at $n = 9$ and 2.64 s at $n = 11$; at $n = 23$ no

[186] `RESEARCH_KOBLITZ_INDEX_CALCULUS.md`.
[187] `autoresearcher:ledger/evidence/EV-FROB-d336b0.yaml`.
[188] `RESEARCH_KOBLITZ_INDEX_CALCULUS.md`.

Table 15: Four-summand compact-orbit index calculus on the ECC2K-130 curve family, single thread, $K = 600$ columns, precompute fully charged (left), and the extrapolation to $n = 131$ ($r = 6.8 \cdot 10^{38}$) at $1.85 \cdot 10^6$ probes/s (right). The minimum base for one expected four-tuple per point is $B > r^{1/4} = 5.1 \cdot 10^9$ points, $K \geq 2 \cdot 10^7$.

n	r (bits)	rank wall	probes/relation
61	47.2	13.1 s	$\sim 4 \cdot 10^4$
71	52.3	1,485 s	$\sim 4.6 \cdot 10^6$
73	56.3	18,398 s	$5.7 \cdot 10^7$

K	probes/relation	total probes	core-years
600	$1.1 \cdot 10^{29}$	$6.6 \cdot 10^{31}$	$1.1 \cdot 10^{18}$
10^6	$4.0 \cdot 10^{22}$	$4.0 \cdot 10^{28}$	$6.8 \cdot 10^{14}$
10^9	$4.0 \cdot 10^{16}$	$4.0 \cdot 10^{25}$	$6.8 \cdot 10^{11}$
10^{12}	$4.0 \cdot 10^{10}$	$4.0 \cdot 10^{22}$	$6.8 \cdot 10^8$

instance finished, one running 25 minutes without returning: roughly $2^{2.3}$ per field bit over that range, from three points.[189]

The compact-orbit ladder. A separate line of work solves rungs of the same curve family ($a = 0$, the ECC2K-130 equation) with a four-summand decomposition, at $n = 41$ and 53 over 85- and 220-column bases and at $n = 61, 71, 73$ over a 600-column base, fully charged, and fits the rank cost. Table 15 is the measured cost and its extrapolation to $n = 131$ under the fit probes/relation $\approx C\, r/(n^2 K^2)$ with $C = 1$.[190] The fitted C (0.33 at $n = 61$, 1.51 at $n = 71$, 1.27 at $n = 73$, recomputed here from the exact subgroup orders) disagrees with the structural model's 0.25 by a factor of 1.3 to 6, and the note records the gap as open; the extrapolation uses $C = 1$, between the structural 0.25 and the fitted 1.3–1.5 of the two largest rungs.

The verdict is in the unit of the rest of the paper. Rho at $2^{60.9}$ iterations on one RTX PRO 6000 is about five GPU-years at the campaign's 14.11 B/s. Four-summand index calculus has total work about $r/(n^2 K)$ probes: at $K = 10^9$ that is $4 \cdot 10^{25}$, some $2 \cdot 10^7$ times rho's iteration count with one probe counted as one step (uncalibrated), and the base at which the precompute alone equals rho is $K \approx 1.8 \cdot 10^{16}$ columns, beyond any linear algebra contemplated. The rungs that beat rho do so only in the online class after a precompute that is excluded from the clock (at $n = 73$, 18,400 s of precompute against 46 s of rho online, a 400× total-work deficit traded for a 185× exploratory online wall ratio on one frozen target, on a shared host), and with the precompute charged the method spends 26–1,200× more probes than one rho solve takes steps on every rung, again uncalibrated. Beating rho on total work needs a relation-shape change (five- or six-summand relations, whose per-point yield scales as $B^5/(120r)$), not engineering, and that oracle is unmeasured.[191] In this repository's boundary ledger the Koblitz rows at a 39-bit subgroup run from $S = \text{ops}/\sqrt{r} = 58.8$ (300× signed-Frobenius rho) for the plain three-summand meet-in-the-middle oracle down to 5.12 (26×; 5.92 and 30× on re-measurement) for a balanced two-summand base; the secondary complete-CPU panel on disjoint public points at $n = 37, 41, 53$ has rho ahead in every cell.[192][193]

10.2 Point decomposition against the per-call budget

The sibling library prices the same question from the other end: given rho's $2^{60.81}$ orbit-walk iterations, how fast must one m-summand decomposition be for index calculus to fit inside that budget, and how fast is the best one measured? With a base of 2^ℓ points, $m = 4$ at $\ell = 29$ needs $2^{48.6}$ decompositions and $2^{60.0}$ of linear algebra, leaving 2^{12} rho-iterations per decomposition; $m = 5$ at $\ell = 28$ leaves 2^{33}; $m = 6$ at $\ell = 24$ leaves 2^{37}; $m = 3$ has no budget at all because

[189] `ecc2k130/README.md, "Index calculus: SAT point decomposition".`
[190] `RESEARCH_ECC2K130_IC_FEASIBILITY.md.`
[191] `RESEARCH_ECC2K130_IC_FEASIBILITY.md, Âğ3-4.`
[192] `research/notes/index-calculus/RESEARCH_IC_BOUNDARY_LEDGER.md.`
[193] `docs/ic/BOUNDARY_TARGETS.md.`

the linear algebra alone exceeds rho. The best measured decomposition is the non-algebraic meet-in-the-middle at exponent $\lceil m/2 \rceil$ in ℓ, and extrapolated to $n = 131$ it is 2^{58}, 2^{84} and 2^{72} for $m = 4, 5, 6$: 35–51 bits over the budget. Algebraic solvers land far higher: SAT fits at 2^4 ($m = 3$) to 2^7 ($m = 4$) per unit of ℓ, msolve grows faster still, and WDSat at about 2^3, the exhaustive-search exponent. An attack needs a decomposition about 2^{35}–2^{50} times faster than meet-in-the-middle at $\ell \approx 25$–30, which is the obstruction that stops Wagner's k-tree from applying to elliptic curves; the trace-homomorphism clause that halves UNSAT conflicts is a constant, not an exponent.[194] A Riemann–Roch (Nagao) oracle on the real curve was measured at factor-base dimensions 4–8 and its derived product law (relations $\times$ targets $\times$ oracle $= m \cdot 2^{131}$) leaves even a free oracle above rho at $m = 3$ ($2^{68.6}$) and below it only from $m = 4$ ($2^{56.4}$, leaving $2^{4.4}$ of headroom for the oracle itself), where no measured oracle comes close.[195]

10.3 What the negatives say

Each alternative was carried to a measurement and each landed above rho by a margin that is a change of exponent, not of constant. The honest summary for ECC2K-130 is that the generic-group floor, $\sqrt{\pi\ell/(2 \cdot 262)}$ class-walk steps, is still the number the campaign is walking toward, and that the only lever that has moved in this work is the cost of one step.

11 Related work

The 2009 attempt. Bailey et al. [3] fixed the iteration function this campaign walks, $R \mapsto \sigma^j(R) + R$ with j from the weight of x, the weight-34 distinguished-point rule, and the $2^{60.9}$ budget, and reported per-platform rates on Core 2 (533 cycles per iteration with 128-bit vectors), Cell, GPUs, FPGAs [14] and ASIC estimates; their SHARCS paper [2] priced the whole Certicom ECC2 list. Bos, Kleinjung, Niederhagen and Schwabe's Cell implementation [9] is the bitsliced lineage this client's CPU and first GPU backends belong to. The type-II optimal-polynomial-basis multiplication of Bernstein and Lange [5] is used verbatim; the conversion networks generated here match their hand-designed chains to within a few instructions (Section 4.1). Bernstein's bit operation counts for binary multiplication [4] are the reference the three-input-logic finding is measured against. Bernstein et al.'s later FPGA designs [7], which set the current size record for a completed binary-curve computation by solving a $2^{117.35}$-order instance on sect113r2 and were extrapolated to a 127-bit field, are the calibration for the pre-board estimate of Section 8, which the measurement then met. Harley's group solved ECC2K-95 in 1998 and ECC2K-108 in April 2000, the latter with about 1,300 volunteers' machines over four months [19, 20]; those two are the only Koblitz challenges completed, and their walks are the ancestors of the one in Section 3.1.

Rho on automorphism classes. The $\sqrt{2m}$ saving from Frobenius and negation on Koblitz curves is due to Wiener and Zuccherato [30] and Gallant, Lambert and Vanstone [17], generalised by Duursma, Gaudry and Morain [13]. Teske's r-adding walks [28] and the Bernstein–Lange–Schwabe treatment of fruitless cycles under the negation map [6] are the background to Section 3.2; the cycle classes measured there are the τ-relation analogue, and the measured walk constants of Table 1 are, to our knowledge, the first reported for this branch distribution on this family. The distinguished-point method is van Oorschot and Wiener's [29]. The 112-bit prime-field record of Bos et al. [10] is the nearest completed computation in scale.

Index calculus on binary curves. Semaev's summation polynomials [27], Gaudry's and Diem's index calculus [12, 18], the Faugère–Perret–Petit–Renault and Petit–Quisquater analy-

[194] `cryptanalysis:experiments/pdp-scaling/README.md`.
[195] `autoresearcher:ledger/evidence/EV-ICPERF-10c5fc.yaml`.

ses of Weil-descent systems [15, 25] and Galbraith and Gebregiyorgis's characteristic-two algorithms [16] are the methods Section 10 prices on this curve. The structural closure at $n = 131$ (no Frobenius-stable subspace of intermediate dimension) is elementary once the normal basis is in view, and the measured ladder confirms that the surviving weight-bounded bases do not reach the budget.

Measurement discipline. The companion paper [1] states the boundary-unit-table-class rule and applies it across rho, baby-step giant-step, kangaroo and prime-field index calculus; this paper is its ECC2K-130 chapter at full length.

12 Conclusion

Four statements survive the accounting.

First, ECC2K-130 is an arithmetic problem at a known exponent, and the exponent did not move. Index calculus on the curve, priced end to end, is 10^7 times rho at the challenge size with four-summand relations; the best measured decomposition is 35–51 bits short of the per-call budget; the Frobenius-stable subspaces that the literature's construction needs do not exist at $n = 131$. The number the campaign walks toward is still $\sqrt{\pi\ell/(2 \cdot 262)}$ class steps.

Second, the cost of a step on a GPU was measured against a floor written first. Sixteen paired rungs took one RTX PRO 6000 from 0.85 to 14.64 billion complete iterations per second with the campaign's σ-walk, later σ-walk builds reached 15.88, a table walk reached 20.08 as a rate, and the roofline of that build puts the logic pipe at 91% and the carry-less unit at 73% of saturation. Twenty-eight on one card is below both pipes for any walk that performs one affine addition per step; thirty is a two-card number. A 17.0 B/s target declared before a build declined a 16.56 B/s result, which is what a declared target is for. The other GPU arithmetic in the repositories, prime-field and binary, is verified and statically accounted to the same standard and, with the exception of F4 elimination on rented cards, has not yet been measured on a device; the paper says so rather than quoting its emulations as rates.

Third, the step count is measured, not assumed. The campaign's σ-walk costs $2^{60.91}$–$2^{60.92}$ iterations, the 2009 estimate within its own precision; the table walk's cheaper constant was paid back several times over by fruitless cycles until the rule was fixed, and its projected 0.81–0.86 cost per solve awaits the paired run that would confirm it. An FPGA engine designed from the same model measured 8.25 G steps/s on one F2 against a pre-board central estimate of 8, level with the GPU on cost per solve.

Fourth, the campaign is a corpus, and the corpus is honest. One report per $2^{28.41}$ iterations at cutoff 32, a figure that replaced a wrong $2^{27.9}$ and says so; 486 million points and $2^{56.68}$ iterations with no collision, 5.4% of the expected work, every record re-walkable from its seed and every rate in this paper re-walked before it was recorded. The fleet that produced most of it was reclaimed on 2026-09-18, the last point landed on 2026-10-07, and collection is paused. Whether the remaining 94.6% is walked is a question of money, not of mathematics; this paper's contribution is to have priced it.

A Reproduction

All commands run from the repository root unless a directory is named. CUDA rows need the card named in the row; the host checks do not.

Field, walk and solver.

```
cd ecc2k130 && make generate && make test        # 52 checks, GF(2^23..131)
make break-small   # planted logs, GF(2^41)
python3 ecc2k130/examples/rho_toy.py   # the walk on GF(2^23)
```

GPU rates (RTX PRO 6000 unless noted).

```
make -C ecc2k130 bench-rtx-pro6000   # audited sigma preset, 14.64 B/s
make -C ecc2k130 audit-rtx-pro6000   # same, with DP34 collection
make -C ecc2k130 gpu-rtx-pro6000-20b   # one-block table walk, 20.08 B/s
make -C ecc2k130 gpu-rtx-pro6000-sigma-fused      # 15.44 B/s
make -C ecc2k130 gpu-b200-19b   # B200 autosweep arm, 19.40 B/s
make -C ecc2k130 bench-local-packed   # RTX PRO 4500 bridge map, 12.50 B/s
python3 ecc2k130/benchmark_instance.py   # concurrent-instance accounting
```

Every rate is a paired comparison; the candidate/control pairs, raw samples and re-walk logs are in the `ecc2k130/benchmarks/` directory each note links.

Other GPU kernels (host emulation; no device needed).

```
make -C gpu/ecc test        # secp256k1 field, folds, BSGS; ptx_asm_check
make -C gpu/btcpuzzle test  # kangaroo invariant, planted 28/32/36-bit solves
make -C gpu/nist test       # P-256/P-384 against Boost.Multiprecision
make -C gpu/ecc2k test      # ECC2K-95 field, steppers, pair table, tsan
make -C gpu/semaev test && make -C gpu/semaev bench-host L=7
```

Walk constant and cycles.

```
make -C ecc2k130 test-cycle-rule   # 19,200 tau-cycles refused
# scatter runs and c measurements: ecc2k130/WALK-CONSTANT.md, section 4 and 11.6
```

FPGA.

```
cd hdl/ecc2k130 && python3 ecc2k_ref.py check && make   # vectors, testbenches
make rate   # 6,144 steps, clk/step
cd aws && NENG=128 CLK_MHZ=333 ./build_afi.sh push && ./build_afi.sh launch
```

Campaign.

```
cat ecc2k130/aws/campaign.json   # frozen configuration
ecc2k130/benchmarks/dp-interval/   # 2^28.41 calibration, raw TSV
ecc2k130/CERTIFICATION.md   # storage and replay protocol
research/ecc2k130_rho_audit_20260925/   # live snapshot and audit
```

Alternatives.

```
cd ecc2k130/codegen && \
  python3 indexcalc.py --m 11 --points 3 --weight 3 --trials 5
RESEARCH_ECC2K130_IC_FEASIBILITY.md   # compact-orbit ladder, n=131
cryptanalysis:experiments/pdp-scaling/   # decomposition budget
```

References

[1] Anonymous. The constant in front of $\sqrt{n}$: measured Pollard rho, ECC2K-130 arithmetic, and index calculus over prime fields. Companion draft, https://github.com/aburan28/cryptanalysis/tree/main/paper, 2026.

[2] Daniel V. Bailey, Brian Baldwin, Lejla Batina, Daniel J. Bernstein, Peter Birkner, Joppe W. Bos, Gauthier van Damme, Giacomo de Meulenaer, Luis Julian Dominguez Perez, Junfeng Fan, Tim Güneysu, Frank Gürkaynak, Thorsten Kleinjung, Tanja Lange, Nele Mentens, Christof Paar, Francesco Regazzoni, Peter Schwabe, and Leif Uhsadel. The Certicom challenges ECC2-X. In *Workshop on Special-purpose Hardware for Attacking Cryptographic Systems – SHARCS 2009*, 2009.

[3] Daniel V. Bailey, Lejla Batina, Daniel J. Bernstein, Peter Birkner, Joppe W. Bos, Hsieh-Chung Chen, Chen-Mou Cheng, Gauthier van Damme, Giacomo de Meulenaer, Luis Julian Dominguez Perez, Junfeng Fan, Tim Güneysu, Frank Gürkaynak, Thorsten Kleinjung, Tanja Lange, Nele Mentens, Ruben Niederhagen, Christof Paar, Francesco Regazzoni, Peter Schwabe, Leif Uhsadel, Anthony Van Herrewege, and Bo-Yin Yang. Breaking ECC2K-130. IACR Cryptology ePrint Archive, Report 2009/541, 2009.

[4] Daniel J. Bernstein. Batch binary Edwards. In *CRYPTO 2009*, pages 317–336. Springer, 2009.

[5] Daniel J. Bernstein and Tanja Lange. Type-II optimal polynomial bases. In *Arithmetic of Finite Fields – WAIFI 2010*, volume 6087 of *LNCS*, pages 41–61. Springer, 2010.

[6] Daniel J. Bernstein, Tanja Lange, and Peter Schwabe. On the correct use of the negation map in the Pollard rho method. In *Public Key Cryptography – PKC 2011*, volume 6571 of *LNCS*, pages 128–146. Springer, 2011.

[7] Daniel J. Bernstein, Susanne Engels, Tanja Lange, Ruben Niederhagen, Christof Paar, Peter Schwabe, and Ralf Zimmermann. Faster elliptic-curve discrete logarithms on FPGAs. IACR Cryptology ePrint Archive, Report 2016/382, 2016.

[8] Eli Biham. A fast new DES implementation in software. In *Fast Software Encryption – FSE 1997*, pages 260–272. Springer, 1997.

[9] Joppe W. Bos, Thorsten Kleinjung, Ruben Niederhagen, and Peter Schwabe. ECC2K-130 on Cell CPUs. In *Progress in Cryptology – AFRICACRYPT 2010*, volume 6055 of *LNCS*, pages 225–242. Springer, 2010.

[10] Joppe W. Bos, Marcelo E. Kaihara, Thorsten Kleinjung, Arjen K. Lenstra, and Peter L. Montgomery. Solving a 112-bit prime elliptic curve discrete logarithm problem on game consoles using sloppy reduction. *International Journal of Applied Cryptography*, 2(3):212–228, 2012.

[11] Certicom Research. The Certicom ECC challenge. https://www.certicom.com/content/certicom/en/the-certicom-ecc-challenge.html, 1997.

[12] Claus Diem. On the discrete logarithm problem in elliptic curves. *Compositio Mathematica*, 147(1):75–104, 2011.

[13] Iwan Duursma, Pierrick Gaudry, and François Morain. Speeding up the discrete log computation on curves with automorphisms. In *ASIACRYPT 1999*, pages 103–121. Springer, 1999.

[14] Junfeng Fan, Daniel V. Bailey, Lejla Batina, Tim Güneysu, Christof Paar, and Ingrid Verbauwhede. Breaking elliptic curve cryptosystems using reconfigurable hardware. In *International Conference on Field Programmable Logic and Applications – FPL 2010*, pages 133–138. IEEE, 2010.

[15] Jean-Charles Faugère, Ludovic Perret, Christophe Petit, and Guénaël Renault. Improving the complexity of index calculus algorithms in elliptic curves over binary fields. In *EUROCRYPT 2012*, pages 27–44. Springer, 2012.

[16] Steven D. Galbraith and Shishay W. Gebregiyorgis. Summation polynomial algorithms for elliptic curves in characteristic two. In *INDOCRYPT 2014*, pages 409–427. Springer, 2014.

[17] Robert P. Gallant, Robert J. Lambert, and Scott A. Vanstone. Improving the parallelized Pollard lambda search on anomalous binary curves. *Mathematics of Computation*, 69(232): 1699–1705, 2000.

[18] Pierrick Gaudry. Index calculus for abelian varieties of small dimension and the elliptic curve discrete logarithm problem. *Journal of Symbolic Computation*, 44(12):1690–1702, 2009.

[19] Robert Harley. Elliptic curve discrete logarithms: ECC2K-95. Announcement of the solution of the Certicom ECC2K-95 challenge, 1998.

[20] Robert Harley, Damien Doligez, Daniel de Rauglaudre, and Xavier Leroy. Elliptic curve discrete logarithms: ECC2K-108. INRIA announcement of the solution of the Certicom ECC2K-108 challenge, 13 April 2000, 2000.

[21] Toshiya Itoh and Shigeo Tsujii. A fast algorithm for computing multiplicative inverses in GF(2^m) using normal bases. *Information and Computation*, 78(3):171–177, 1988.

[22] Anatolii Karatsuba and Yuri Ofman. Multiplication of many-digital numbers by automatic computers. *Doklady Akademii Nauk SSSR*, 145:293–294, 1962.

[23] Neal Koblitz. CM-curves with good cryptographic properties. In *CRYPTO 1991*, pages 279–287. Springer, 1992.

[24] Peter L. Montgomery. Speeding the Pollard and elliptic curve methods of factorization. *Mathematics of Computation*, 48(177):243–264, 1987.

[25] Christophe Petit and Jean-Jacques Quisquater. On polynomial systems arising from a Weil descent. In *ASIACRYPT 2012*, pages 451–466. Springer, 2012.

[26] J. M. Pollard. Monte Carlo methods for index computation (mod p). *Mathematics of Computation*, 32(143):918–924, 1978.

[27] Igor Semaev. Summation polynomials and the discrete logarithm problem on elliptic curves. IACR Cryptology ePrint Archive, Report 2004/031, 2004.

[28] Edlyn Teske. On random walks for Pollard's rho method. *Mathematics of Computation*, 70(234):809–825, 2001.

[29] Paul C. van Oorschot and Michael J. Wiener. Parallel collision search with cryptanalytic applications. *Journal of Cryptology*, 12(1):1–28, 1999.

[30] Michael J. Wiener and Robert J. Zuccherato. Faster attacks on elliptic curve cryptosystems. In *Selected Areas in Cryptography – SAC 1998*, pages 190–200. Springer, 1999.